



Exchange Rate Volatility and Foreign Direct Investment in Egypt: Evidence from an Exponential Generalized Autoregressive Conditional Heteroskedasticity–Autoregressive Distributed Lag Analysis, 2001–2024

Emad Said Khalil Abdalla^{*}, Ashraf Salah Eldin Saleh, Marwa Elsherif

Department of Finance, College of Management and Technology, Arab Academy for Science, Technology and Maritime Transportation, 21913 Alexandria, Egypt

^{*} Correspondence: Emad Said Khalil Abdalla (Eabdalla@tmg.com.eg)

Received: 02-08-2026

Revised: 04-07-2026

Accepted: 04-22-2026

Citation: Abdalla, E. S. K., Saleh, A. S. E., & Elsherif, M. (2026). Exchange rate volatility and foreign direct investment in Egypt: Evidence from an Exponential Generalized Autoregressive Conditional Heteroskedasticity–Autoregressive Distributed Lag analysis, 2001–2024. *J. Account. Fin. Audit. Stud.*, 12(2), 105–129. <https://doi.org/10.56578/jafas120203>.



© 2026 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited under the CC BY 4.0 license.

Abstract: The relationship between exchange rate volatility and foreign direct investment (FDI) inflows in emerging economies has remained a central issue in international finance, particularly in economies exposed to macroeconomic instability and external shocks. In this study, the impact of real exchange rate volatility (VOLREXR) on FDI inflows in Egypt was examined using quarterly data spanning the period from 2001 to 2024. Exchange rate volatility was first estimated through the Exponential Generalized Autoregressive Conditional Heteroskedasticity (EGARCH) model in order to capture asymmetric responses to exchange rate shocks and to generate a more accurate measure of exchange rate uncertainty. Subsequently, the dynamic relationship between exchange rate volatility and FDI was investigated within the Autoregressive Distributed Lag (ARDL) framework, which enabled the estimation of both short-run adjustments and long-run equilibrium effects in the presence of mixed orders of integration. The empirical analysis was supported by unit root testing, bounds cointegration testing, and an error correction specification, while model adequacy and robustness were verified through a comprehensive set of diagnostic and stability tests, including serial correlation, heteroskedasticity, normality, multicollinearity, and structural stability assessments. The results indicate that VOLREXR exerts a statistically significant and negative effect on aggregate FDI inflows in Egypt in both the short run and the long run, suggesting that heightened exchange rate uncertainty weakens investor confidence and discourages capital commitment. Among the control variables, market size, represented by gross domestic product (GDP), was found to exert a positive and statistically significant influence on foreign investment, confirming the importance of domestic economic expansion in attracting international capital. By contrast, inflation and external debt were found to impose adverse effects on investment performance, reflecting the destabilizing consequences of macroeconomic imbalances. Market capitalization, however, was shown to contribute positively to FDI inflows, highlighting the role of financial market development in strengthening investment attractiveness. Overall, the findings underscore the importance of exchange rate stability and coherent macro-financial policy coordination in fostering a predictable investment climate and supporting sustainable long-term capital inflows into the Egyptian economy.

Keywords: Exchange rate volatility; Foreign direct investment; Egypt; EGARCH model; ARDL model; Cointegration analysis; Macroeconomic stability; Financial market development

1. Introduction

Foreign direct investment (FDI) has become one of the most important drivers of economic growth in developing countries, serving not only as a source of external capital inflow to the host countries but also as a channel for knowledge and technology transfer beyond providing financial resources. FDI facilitates the transfer of innovation and managerial expertise, contributing to sustainable economic growth (Elmoghbari, 2025). Moreover,

FDI serves as a consistent source of foreign exchange, enabling central banks to maintain healthy reserve levels, thereby enhancing financial stability.

Exchange rate volatility is a fundamental financial macroeconomic variable influencing the dynamics of FDI, and its significance has grown among policymakers in recent decades, particularly as an increasing number of countries have transitioned to floating exchange rate regimes (Elmoghlar, 2025). Exchange rate instability, characterized by unpredictable fluctuations, creates significant challenges for macroeconomic and financial stability, particularly in developing economies like Egypt that are highly integrated into the global system. Such volatility generates considerable uncertainty within financial markets, thereby affecting capital flows, altering asset valuations, and disrupting the broader equilibrium of financial and economic activity.

Several studies, including Alnaa & Ahiakpor (2020); Elbadry & Mandour (2020); El-Sayed (2020); Ezekiel & Temidayo (2023); Fatima et al. (2023); Jamal & Bhat (2022); Kaya & Erden (2022); Moraghen et al. (2021); Nguyen et al. (2022); Oladeji & Musa (2022); Nadine et al. (2021); and Shafique et al. (2022), have investigated the relationship between exchange rate volatility and FDI. Empirical findings indicate that exchange rate volatility negatively affects FDI. However, some scholars argue that exchange rate depreciation, by generating volatility, can unexpectedly stimulate greater inflows of FDI into host countries. Meanwhile, some studies emphasize the negative effects of volatility on FDI, like Elbadry & Mandour (2020); Fatima et al. (2023); Jamal & Bhat (2022); Moraghen et al. (2021); Nguyen et al. (2022). While others report a positive relationship between exchange rate volatility and FDI flows, El-Sayed (2020); Ezekiel & Temidayo (2023), Kaya & Erden (2022); Nadine et al. (2021), Shafique et al. (2022). Yet few studies report no significant relationship among the variables.

Egyptian literature on FDI has produced mixed results: some studies find that FDI has a negative relationship with volatility and a positive relationship with Egyptian pound depreciation, whereas a few studies find the relationship insignificant. In this instance, Elbadry & Mandour (2020) supported the negative effect of exchange rate volatility on FDI. Using Autoregressive Distributed Lag (ARDL) cointegration and Granger causality for Egypt (1990–2018), the evidence suggests that volatility, rather than the level of the exchange rate itself, weakens investor confidence and discourages sustained foreign investment. This result is inconsistent with the study by Fadl & Ghoneim (2020), which used least squares for Egypt (1990–2015); they found a significant positive effect of exchange rate depreciation on FDI. They attributed this to government devaluation policies, which made Egypt more attractive to foreign investors. They also highlighted other positive factors such as trade openness, gross domestic product (GDP) per capita growth, urbanization, and natural resource endowment. In addition, Abdelgany (2020), covering 1980–2018 with ARDL, concluded that depreciation of the Egyptian pound encouraged FDI, with significant short- and long-run effects recommending policies to boost growth, infrastructure, and job creation. Also, Al-Sayed (2020), using ARDL for 1977–2017, found a positive and significant long-run relationship between the nominal exchange rate and FDI, alongside other variables such as GDP growth, trade openness, reserves, inflation, and external debt (EXDBTG). El-Sayed (2020), also covering 1977–2017 with ARDL, supported the long-run positive effect of exchange rate depreciation on FDI. Most Egyptian studies highlight the positive impact of exchange rate devaluation on FDI; Hend et al. (2021) found that devaluation attracted FDI but deterred indirect investment, stressing the importance of aligning fiscal and monetary policies with political stability. Nadine et al. (2021), however, revealed that exchange rate volatility had a positive effect on FDI, alongside factors such as market size, trade, and labor productivity. Al-Ghiash (2022) proposed an optimal devaluation range (42.5–92.5 EGP/USD) to attract FDI, recommending continuation of Egypt's floating exchange rate policy. The literature suggests that while volatility may sometimes encourage investment under certain conditions, controlled depreciation consistently attracts FDI, especially when supported by sound macroeconomic and structural policies.

Recent evidence further illustrates the complexity of the relationship between exchange rate volatility and FDI in Egypt. Asamoah et al. (2022), examining 40 African countries, including Egypt, for 1990–2018 using Generalized Method of Moments (GMM), found that volatility discourages FDI. They concluded that financial development is essential to attract investment and recommended stabilizing exchange rates through growth in manufacturing and industry. Similarly, Elagouza (2023), using a multivariate Vector Error Correction (VEC) model for 2000–2018, argued that exchange rate liberalization and devaluation policies negatively affected the Egyptian economy. The findings suggest that depreciation deterred FDI rather than encouraging it, highlighting the risks of excessive reliance on currency adjustments. More recent work by Marwa et al. (2024) confirmed significant negative effects of exchange rate fluctuations on FDI, emphasizing the importance of stability and sound macroeconomic governance. In contrast, Elmoghlar (2025) further suggested that devaluation positively influences FDI attraction, though other macroeconomic factors showed mixed significance. Meanwhile, Abdallah (2023) offered a different perspective, finding no significant relationship between exchange rate movements and FDI in Egypt during 2000–2022. Using Autoregressive Conditional Heteroskedasticity (ARCH) and Granger causality tests, the study concluded that despite depreciation and volatility, FDI inflows were unaffected, suggesting that other structural and institutional factors may play a more decisive role.

Furthermore, several international studies also produced mixed results; especially, these studies emphasize that volatility discourages FDI. Moraghen et al. (2021) demonstrated that exchange rate instability significantly reduces

investment inflows, highlighting investor risk aversion. Their findings suggest that while depreciation may encourage FDI, volatility diminishes investor confidence and deters long-term commitments, and the volatility consistently emerges as a barrier to FDI, reinforcing the importance of stable exchange rate regimes to sustain investor trust. Other studies highlight the benefits of depreciation. Frenzel Baudisch (2018), analyzing 66 economies between 1995 and 2015, found that real depreciation increased FDI in manufacturing but reduced it in services, with no significant effect on the primary sector. This sector-specific evidence suggests that depreciation can attract investment, particularly in industries where cost competitiveness is critical. In addition, Hniya et al. (2021) found that exchange rate volatility reduced FDI in both the short and long run in Tunisia (1980–2018), emphasizing the importance of time-specific volatility. Nguyen et al. (2022), analyzing 156 economies (2002–2017), also reported a negative impact, recommending policies to reduce uncertainty and improve business environments. Ozigbo & Anuya (2023) highlighted similar short-term adverse effects in Nigeria, while Fatima et al. (2023) confirmed negative impacts in India. Abdel Aziz & Al-Ajrawi (2023) found a statistically significant short-run negative relationship in Iraq's agriculture sector, though no long-run effect was observed. Contrasting evidence shows that volatility can sometimes encourage FDI. Shafique et al. (2022) found that long-run volatility positively impacted FDI in Pakistan, driven by depreciation. Kaya & Erden (2022) similarly reported positive volatility-FDI effects across 16 emerging markets, supporting the idea that depreciation attracts investment. Which means in certain contexts, depreciation and volatility may signal opportunities for investors, particularly in emerging markets where cost competitiveness is a key driver. Other studies highlight sectoral or country-specific differences. El Rhadbane & El Moudden (2022) found volatility had a negative impact in Morocco, a positive impact in Turkey, and no significant effect in Egypt, supporting the role of structural and policy differences. Lin & Chen (2022), analyzing Taiwan, reported that real exchange rate (RER) levels positively influenced FDI, while volatility effects varied depending on investor motives and market competition. Udoinyang & Udoinyang (2024) reported a significant negative impact of volatility on FDI in Nigeria (1986–2021), both in the short and long run, using ARDL and Generalized Autoregressive Conditional Heteroskedasticity (GARCH), and recommended comprehensive policy measures beyond exchange rate dynamics. Similarly, Sultana et al. (2024) found that exchange rates negatively affect FDI in the short run across 11 emerging economies, though positively in the long run, highlighting time-dependent dynamics. Zaharum et al. (2024), analyzing Malaysia (1992–2021), found positive effects of exchange rate levels, GDP growth, and moderate inflation on FDI inflows, concluding that stabilizing exchange rates and aligning policies with broader economic goals enhance the investment climate. Sultana et al. (2024) also noted that depreciation can attract FDI in the long run, supporting the idea that flexible regimes may encourage investment. Joel & Jeffrey (2023), examining Nigeria (1986–2021) with Exponential Generalized Autoregressive Conditional Heteroskedasticity (EGARCH) and GMM, found no statistically significant impact of volatility on FDI inflows despite persistent depreciation, recommending stronger exchange rate management. Kalu (2024) similarly reported no significant relationship between volatility and FDI in Nigeria (1981–2018). In addition, Nduka et al. (2024) found that fluctuations in exchange rates, including official exchange rate, RER, and nominal exchange rate, do not significantly impact FDI inflows. The study underscores the need for Policymakers to diversify investment promotion strategies, fortify economic fundamentals, engage in targeted marketing, and consider implementing hedging mechanisms to alleviate currency risk for potential investors.

Recent empirical studies have demonstrated that exchange rate volatility exerts an adverse influence on FDI. Imamboccus et al. (2024), analyzing Mauritius tourism (1999–2019), found that exchange rate volatility negatively affected tourism demand in the short run, with mixed and mostly insignificant effects across other markets. Faizi et al. (2024), examining India (1990–2020), reported that volatility significantly deterred FDI inflows in the short run, though not in the long run, stressing the importance of stabilizing short-term fluctuations. Shabbir et al. (2025) similarly found that volatility discouraged foreign investment in Pakistan (1991–2024), recommending tax reforms and exchange rate stabilization to strengthen investor confidence. Lajevardi & Chowdhury (2024) found that real effective exchange rate (REER) levels and volatility positively influenced FDI inflows in Canada (2007–2022), with sectoral differences showing stronger long-run effects in extractive and manufacturing industries. Their study emphasized the importance of sector-specific policy strategies to manage exchange rate impacts. In contrast, Ben-Obi et al. (2025), analyzing Nigeria (1995–2022) with Nonlinear Autoregressive Distributed Lag (NARDL), found no significant long-run asymmetric impact of volatility on FDI, though volatility directly influenced inflation. The study recommended improving the ease of doing business and strengthening security infrastructure to attract investment.

These inconsistencies in previous studies' results are often attributed to the methodological differences, the choice of volatility-measurement models from the GARCH family, utilizing different sophisticated econometric techniques (ARDL, NARDL, Vector Autoregression (VAR), Vector Error Correction Model (VCEM), GMM, etc.), and model specifications that can yield different results, and included control variables (GDP, interest rates) demonstrably shape results; varying sample sizes and diverse time periods also contribute to unreliable conclusions for computing volatility and inadequate analysis of the real exchange rate volatility (VOLREXR) effects on FDI, especially in the majority of previous Egyptian studies, which masks the study results.

The present study seeks to add meaningful contributions to literature; unlike many of the previous studies, it

improves upon earlier methodological limitations by incorporating a more adequate treatment of VOLREXR because the analysis considers both short-run and long-run fluctuations, recognizing that the effects of volatility on FDI may vary over different time horizons. To strengthen the robustness of the findings, several diagnostic procedures are applied to mitigate endogeneity concerns and ensure that the estimated models provide reliable and valid results. These insights can serve as a valuable resource for both policymakers and investors, offering evidence-based guidance for designing targeted strategies to attract and sustain FDI, ultimately enhancing Egypt’s competitiveness in the global marketplace. In addition, the contribution of the research in creating more accurate customized models for better predicting future movements extends the literature database of the FDI determinants and provides better insights to academics and policymakers for better understanding the relationship to support the economic and financial stability.

In addition, this study addresses the central problem of the continuity of VOLREXR and its implications on FDI in Egypt during the period 2001–2024, using Central Bank of Egypt (CBE) quarterly data. This period was marked by recurrent economic and geopolitical shocks because Egypt’s economy from 1998 to 2024 faced repeated shocks, terrorism, global crises, revolutions, and International Monetary Fund (IMF)-driven devaluations, culminating in a severe currency collapse with the pound floating at 49.5 per USD in March 2024. FDI inflows fluctuated, turning negative in 2011, recovering gradually, then rebounding post-COVID to USD 10.04 billion in 2022. A historic surge in 2023/24 lifted inflows to USD 46.06 billion, dominated by the Ras El-Hekma project but concentrated in low-value-added sectors (Figure 1 shows the official exchange rate of Egypt, 1998–2024), and (Figure 2 and Table 1) show the FDI inflow and net inflow in Egypt from 2001 to 2024. Building on these insights, the research formulates hypotheses that differentiate between short-run and long-run dynamic responses to VOLREXR. Furthermore, the role of control variables such as market size, GDP growth, inflation, market capitalization (MCAP), and EXDBTG is examined to assess their moderating effects on FDI performance across time horizons.

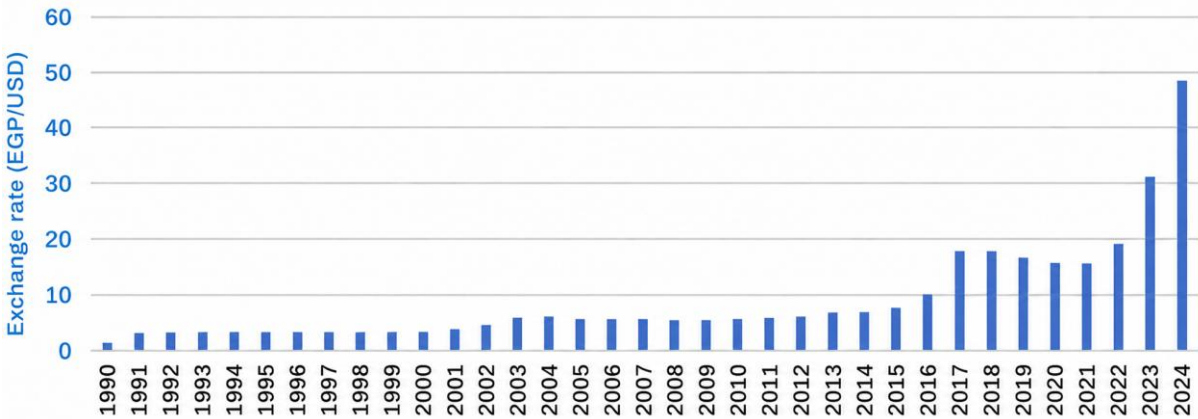


Figure 1. Official exchange rate Egypt 1990–2024
 Note: Sources prepared by researchers based on World Bank statistics.

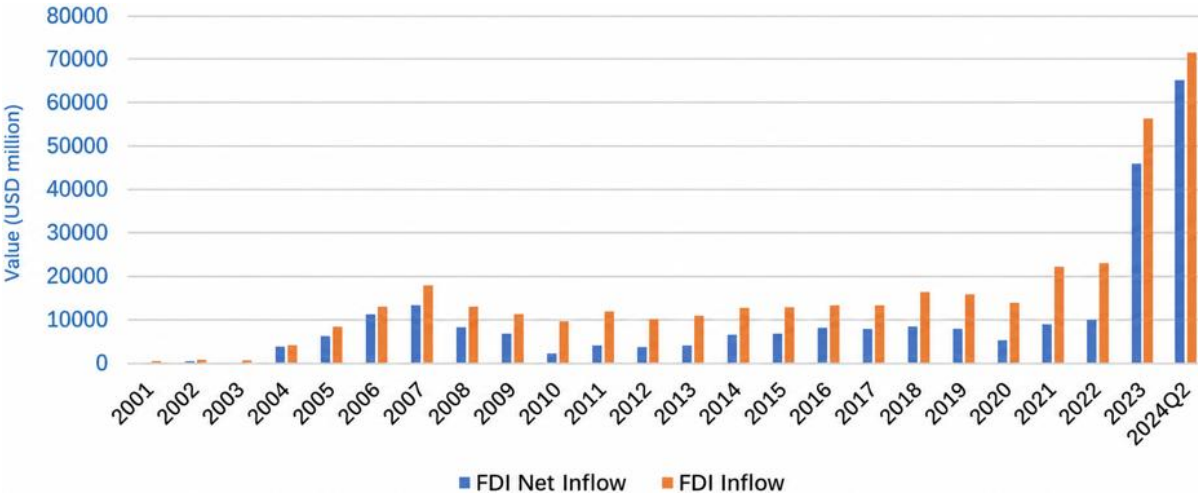


Figure 2. Foreign direct investment (FDI) inflow and net inflow Egypt, 2001–2024Q2
 Note: Central Bank of Egypt (CBE) reports prepared by researchers in 2025.

Table 1. Foreign direct investment (FDI) Inflow and net inflow in Egypt from 2001–2024

Year	FDI Net Inflow (USD Million)	Year	FDI Net Inflow (USD Million)	Year	FDI Inflow (USD Million)	Year	FDI Inflow (USD Million)
2001	428.2	2014	6,379.8	2001	578.7	2014	12,546.2
2002	700.6	2015	6,932.6	2002	945.8	2015	12,528.7
2003	407.2	2016	7,932.8	2003	692.3	2016	13,366.1
2004	3,901.8	2017	7,719.5	2004	4,134.5	2017	13,163.1
2005	6,111.4	2018	8,236.3	2005	8,250.4	2018	16,393.5
2006	11,053.2	2019	7,453.0	2006	13,084.3	2019	15,836.6
2007	13,236.5	2020	5,214.2	2007	17,802.2	2020	13,914.8
2008	8,113.4	2021	8,937.4	2008	12,836.1	2021	22,205.5
2009	6,758.2	2022	10,038.6	2009	11,008.1	2022	23,053.0
2010	2,188.6	2023	46,064.5	2010	9,574.4	2023	56,654.0
2011	3,982.2	2024Q2	65,200.0	2011	11,768.1	2024Q2	71,720.0
2012	3,753.3	-	-	2012	10,273.6	-	-
2013	4,178.2	-	-	2013	10,855.8	-	-

Note: Central Bank of Egypt (CBE) reports prepared by researchers in 2025. “-” indicates not applicable.

Finally, the study seeks to explore a developed understanding of how VOLREXR affects FDI composition, providing empirical evidence that can inform the financial, macroeconomic policy, and investment strategies in Egypt using the EGARCH and ARDL econometric approaches, and also builds on existing literature. The study is designed to answer the following questions: (1) How does VOLREXR impact the inflows of aggregate FDI into Egypt in the short and long run during the period 2001–2024? And (2) What policy implications can be recommended to enhance FDI inflows and strengthen exchange rate stability in Egypt, based on the empirical findings of this study?

2. Methodology

2.1 Data Collection

This study adopts a descriptive and analytical approach, relying exclusively on secondary time-series data. The dataset was obtained from the CBE, including economic reports and external position statements. The dataset is a time series that consists of 94 quarterly observations covering the period 2001–2024. The data were transformed and measured using logarithms (log) during the econometric analysis. Such a dataset provides a comprehensive overview of FDI dynamics in Egypt, capturing both historical fluctuations and recent developments. Table 2 explores the description of research variables. Whereas i = country index dataset and t = time period (year, quarter).

Table 2. Description of research variables

Variables	Description	Unit of Measurement	Source of Data
$FDI_{i,t}$	Foreign direct investment (DV)	Aggregate FDI.	
$VOLREXR_{i,t}$	Real exchange rate volatility (IV)	EGARCH to generate the real exchange rate volatility series.	
$GDP_{i,t}$	Gross domestic product (CV)	The GDP is a widely used and robust measure of market size.	
$CPI_{i,t}$	Consumer price index (CV)	CPI is a key macroeconomic indicator used to measure inflation by tracking changes in the average price level of a basket of goods and services consumed by households.	Central Bank of Egypt (CBE)
$MCAP_{i,t}$	Market capitalization (CV)	Efficient financial markets with higher capitalization, lower transaction costs, enhanced stability, and stronger foreign investor confidence. Each of these dimensions can be expressed using specific measurement units or indicators.	
$EXDBTG_{i,t}$	External debt (CV)	Total External debt is indeed a critical indicator of financial and macroeconomic stability, especially in the context of attracting FDI. Excessive debt burdens increase uncertainty, discourage capital inflows, and weaken long-term investor confidence.	

Note: i denotes the country index and t denotes the time period. DV = dependent variable; IV = independent variable; CV = control variable; EGARCH = Exponential Generalized Autoregressive Conditional Heteroskedasticity.

2.2 Data Analysis

To investigate the impact of VOLREXR on FDI in Egypt, this study employs time-series estimation techniques to minimize bias and enhance the robustness of econometric models. The inclusion of key macroeconomic indicators reflects their established role in guiding investor assessments of economic performance and financial stability, thereby influencing investment decisions Kiliçarslan (2018). The study starts with modeling the VOLREXR using the EGARCH framework introduced by Nelson (1991), which is particularly effective in capturing asymmetric responses to market shocks where positive and negative news exerts differing influences on future volatility and in overcoming the limitations inherent in symmetric volatility models Elbadry & Mandour, (2020). In addition to EGARCH, the study employs the ARDL models to examine cointegration relationships, which is supported by recent empirical literature, including Abdel Aziz & Al-Ajrawi (2023), Abdelgany (2020), Akinlo & Onatunji (2020), Akinlo & Onatunji (2021), Al-Sayed (2020), Asamoah et al. (2022), El Rhadbane & El Mouddeh (2022), Elbadry & Mandour (2020), Elmoghlar (2025), Ezekiel & Temidayo (2023), Hniya et al. (2021), Imamboccus et al. (2024), Jamal & Bhat (2022), Lajevardi & Chowdhury (2024), Marwa et al. (2024), Qamruzzaman et al. (2021), and Sultana et al. (2024), highlighting their efficiency in analyzing the dynamic properties of macroeconomic and financial time-series data, especially in contexts characterized by structural shifts and volatility clustering, particularly the flexibility with variables of differing integration orders, simultaneous estimation of short-run dynamics and long-run equilibrium, and robustness in smaller samples. The study transformed the dependent, independent, and control variables FDI, VOLREXR, GDP, consumer price index (CPI), MCAP, and EXDBTG (into logarithmic form) to improve statistical properties, interpretability, and robustness of results. Log transformation converts them into linear relationships, simplifying estimation. In addition, taking logs reduces scale differences and stabilizes variance, making regression assumptions more valid. To ensure the reliability and validity of the regression estimates, a series of diagnostic tests was conducted. Specifically, the Durbin–Watson d-statistic and the Breusch–Pagan–Godfrey autocorrelation tests, which are applied to the model's residuals to assess whether they are serially correlated. The white heteroscedasticity test, the Jarque–Bera normality test, the variance inflation factor (VIF) to test multicollinearity, and the structural stability by cumulative sum (CUSUM) and CUSUM of square stability tests.

2.3 Testing Stationarity by Augmented Dickey-Fuller and Phillips-Perron Tests

To ensure the validity of econometric estimations, the study first assesses the stationarity of the time series data using the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests. These tests are widely recognized in time series econometrics for evaluating the presence of unit roots and confirming the stability of data over time. Employing both ADF and PP tests enhances the robustness of the stationarity assessment, as convergent results from both methods increase confidence in the reliability of the findings Kiliçarslan (2018). The ARDL model is particularly advantageous in this context due to its flexibility in handling variables that are integrated of order zero $I(0)$ or order one $I(1)$. This feature allows the ARDL framework to be applied even when the underlying variables exhibit mixed levels of integration, thereby avoiding the limitations of traditional cointegration techniques that require uniform stationarity, and ensuring stationarity before model estimation is essential to avoid misleading regression results and to maintain the integrity of the empirical analysis (Asamoah et al., 2022; Elbadry & Mandour, 2020).

2.4 Modelling Exchange Rate Volatility Series by Exponential Generalized Autoregressive Conditional Heteroskedasticity

The EGARCH model, introduced by Nelson (1991), has become a preferred tool in most of the recent empirical research on exchange rate behavior. Its methodological strength lies in its ability to incorporate historical volatility patterns to predict future fluctuations, offering a precise approach to volatility estimation that exceeds symmetric GARCH models in capturing real-world financial dynamics. Abdallah (2023), Adokwe et al. (2019), Alnaa & Ahiakpor (2020), Asmae & Ahmed (2019), Balaban et al. (2019), El Rhadbane & El Mouddeh (2022), Jannat (2020), Joel & Jeffrey (2023), Jokosenumi & Adesete (2018), Latief & Lefen (2018), Lin & Chen (2022), Mackton et al. (2018), Nadine et al. (2021), Shafique et al. (2022), Thujiyanthan (2021), and Udoinyang & Udoinyang (2024) utilized the EGARCH to generate the volatility series to estimate conditional variance and prediction of future VOLREXR due to its effectiveness in modeling conditional variance and forecasting volatility.

2.5 Exponential Generalized Autoregressive Conditional Heteroskedasticity Model Specifications

The EGARCH process involves modeling the logarithm of the conditional variance, generating a robust volatility series that accurately reflects the asymmetric nature of market responses to shocks. The derived volatility series serves as a key explanatory variable in the main regression analysis investigating its impact on investment

flows. The conditional variance for the EGARCH (p, q) model is specified as follows in Eq. (1):

$$\log(h_t) = \varphi + \sum_{i=1}^q \eta_i \left| \frac{u_{t-i}}{\sqrt{h_{t-i}}} \right| + \sum_{i=1}^q \lambda_i \frac{u_{t-i}}{\sqrt{h_{t-i}}} + \sum_{k=1}^q \theta_k \log(h_{t-k}) \quad (1)$$

where,

- LHS is the log of the variance series (h_t), which makes the leverage effect exponential rather than quadratic. This ensures that the estimates are non-negative.
- φ = Constant, η = ARCH effects, λ = asymmetric effects and θ = GARCH effects.
- If $\lambda_1 = \lambda_2 = \dots = 0$, the model is symmetric.
- But if $\lambda_1 < 0$, it implies that bad news (negative shocks) generates larger volatility than good news (positive shocks).

2.6 Autoregressive Distributed Lag Model Specification

The study employs the ARDL model on quarterly data spanning 2001–2024 (94 observations). The ARDL bounds testing approach is particularly suitable for small sample sizes (Muhammad & Abdullahi, 2020), and allows for variables with mixed integration orders, specifically I(0) and I(1), without requiring uniform stationarity. Moreover, the bounds test helps mitigate issues of serial correlation and endogeneity (Rahman & Kashem, 2017). The generalized form of the model specifies FDI as the dependent variable, with VOLREXR as the main independent variable, and GDP, inflation, CPI, MCAP, and EXDBTG as control variables. Following Ambala & Amewu (2022), the linear ARDL framework incorporates appropriate lags of both the dependent and explanatory variables, ensuring dynamic adjustment and robust inference.

$$\text{FDI (DV)} = f(\text{VOLREXR})(\text{IDV}), \text{GDP, CPI, MCAP, EXDBTG (Control variables)} \quad (2)$$

Whereas the dependent FDI is foreign direct investment, the independent variable VOLREXR is real exchange rate volatility; meanwhile, the control variables are the market size, GDP, inflation, CPI, MCAP, and EXDBTG.

Ambala & Amewu (2022) correctly specify the linear ARDL model; the lags of both the dependent variable and control variables must be included. For p lags of our dependent variable, VOLREXR, and k lags of control variables, the research constructed the following ARDL Eq. (3):

$$\text{FDI}_t = \beta_0 + \sum_{i=1}^p \beta_i \text{FDI}_{t-i} + \sum_{i=0}^{K_1} \gamma_i \text{VOLREXR}_{t-i} + \sum_{i=0}^{K_2} \delta_i \text{GDP}_{t-i} + \sum_{i=0}^{K_3} \delta_i \text{CPI}_{t-i} + \sum_{i=0}^{K_4} \rho_i \text{MCAP}_{t-i} + \sum_{i=0}^{K_5} \tau_i \text{EXDBTG}_{t-i} + \varepsilon_t \quad (3)$$

Here, kq for $q = 1, 2, 3, \dots, 5$ denotes the maximum number of lags for VOLREXR, GDP, CPI, MCAP, and EXDBTG, respectively. This research uses EViews 12 software, and the optimal lag for each variable was automatically selected by the software. The ARDL bound test is formulated as follows in Eq. (4):

$$\Delta \ln \text{FDI}_t = \beta_0 + \sum_{i=1}^p \beta_{i\Delta} \ln \text{FDI}_{t-i} + \sum_{i=0}^{K_1} \gamma_{i\Delta} \ln \text{VOLREXR}_{t-i} + \sum_{i=0}^{K_2} \delta_{i\Delta} \ln \text{GDP}_{t-i} + \sum_{i=0}^{K_3} \delta_{i\Delta} \ln \text{CPI}_{t-i} + \sum_{i=0}^{K_4} \rho_{i\Delta} \ln \text{MCAP}_{t-i} + \sum_{i=0}^{K_5} \tau_{i\Delta} \ln \text{EXDBTG}_{t-i} + \lambda_1 \ln \text{GDP}_{t-1} + \lambda_2 \ln \text{CPI}_{t-1} + \lambda_3 \ln \text{MCAP}_{t-1} + \lambda_4 \ln \text{EXDBTG}_{t-1} + \varepsilon_t \quad (4)$$

Δ is the difference operator, and $\ln$ is the natural log of the variables. From Eq. (4), Short-run dynamics are captured by λ_i , for $i = 1, 2, 3, \dots, 5$. and the long-run dynamics are captured by $\beta_i, \gamma_i, \delta_i, \rho_i, \tau_i, \sigma_i$, for $i = 1, 2, 3, \dots, p$. Using a shorter and an error correction model (ECM), Eq. (5) could be written as follows:

$$\Delta \ln \text{FDI}_t = \beta_0 + \sum_{i=1}^p \beta_{i\Delta} \ln \text{FDI}_{t-i} + \sum_{i=0}^{K_1} \gamma_{i\Delta} \ln \text{VOLREXR}_{t-i} + \sum_{i=0}^{K_2} \delta_{i\Delta} \ln \text{GDP}_{t-i} + \sum_{i=0}^{K_3} \delta_{i\Delta} \ln \text{CPI}_{t-i} + \sum_{i=0}^{K_4} \rho_{i\Delta} \ln \text{MCAP}_{t-i} + \sum_{i=0}^{K_5} \tau_{i\Delta} \ln \text{EXDBTG}_{t-i} + \lambda \text{ECT}_{t-1} + \varepsilon_t \quad (5)$$

ECT is the error-correction term that captures the long-run relationship between the variables, and its coefficient measures the speed of adjustment to the long-run equilibrium following any shock to the system.

2.7 Formulating Real Exchange Rate Volatility Series (2001–2024)

The research explored in part three the Eq. (6) used in modelling the volatility by GARCH and exponential EGARCH, and also the data used, quarterly returns of exchange rates on the Egyptian pound, in generating the

real exchange rate volatility (VOLREXR) series. As in most empirical literature, the variable to be modelled is the percentage daily exchange rate return, which is the first difference of the natural logarithm of the exchange rate and is given by the following equation: Abdalla (2012):

$$\text{VOLREXR}_t = 100 * \log \left(\frac{\text{RER}_t}{\text{RER}_{t-1}} \right) \quad (6)$$

where, VOLREXR_t is the quarterly percentage return of the exchange rate; RER_t and RER_{t-1} denote the exchange rate at the current quarter and the previous quarter, respectively.

2.8 Augmented Dickey-Fuller Stationarity Test for Real Exchange Rate Series

The ADF test, as shown in Table 3, the test results for the VOLREXR indicate that the series is stationary at level (0) with no trend and intercept, and the following table summarizes these results as follows.

Table 3. EViews output ADF stationary test for real exchange rate

Test Item	t-Statistic	p-Value*
Augmented Dickey-Fuller test statistic	-7.731098	<0.001
Critical value (1%)	-2.590340	-
Critical value (5%)	-1.944364	-
Critical value (10%)	-1.614441	-

Note: ADF = Augmented Dickey-Fuller. “-” indicates not applicable. * = significant at the 10% level.

The ADF statistic of -7.731 is markedly lower than the critical values at the 1%, 5%, and 10% levels (-2.590 , -1.944 , and 1.614), the associated p -value of 0.0000 , which is well below the 1% significance level, provides strong statistical evidence against the null hypothesis and confirms the suitability of the series for regression analysis, meaning that the VOLREXR series is stationary at level (0), requiring no differencing, and exhibits a constant mean and variance over time, thereby excluding the presence of a unit root. Establishing stationarity is validated for the dataset for econometric applications in both short-run and long-run estimations, which is particularly critical for volatility modeling and robust time-series regression analysis.

2.9 Testing Volatility Returns by the Exponential Generalized Autoregressive Conditional Heteroskedasticity Model

The EGARCH in Table 4 shows that the estimation results indicate a pattern characterized by relatively low long-run variance, pronounced persistence in volatility, an unusual weakening effect in response to large shocks, and evidence of reverse-leverage asymmetry. These dynamics appear to be shaped, at least in part, by CBE’s exchange rate management strategies. Such findings are consistent with earlier empirical studies, which demonstrate that financial time series frequently display persistent volatility and asymmetric responses to shocks (Bollerslev, 1986; Nelson, 1991).

Table 4. EViews12 output modelling exchange rate volatility

Mean Equation				
Variable	Coefficient	Std. Error	z-Statistic	p-Value
C	0.028383	0.002119	13.39510	<0.001
VOLREXR (-1)	0.091670	0.030964	2.960550	0.0031
Variance Equation				
Variable	Coefficient	Std. Error	z-Statistic	p-Value
C (3)	-0.072203	0.026973	-2.676875	0.0074
C (4)	-0.564960	0.071967	-7.850217	<0.001
C (5)	0.424394	0.044146	9.613446	<0.001
C (6)	0.907589	0.00012	7700	<0.001

Note: “C” terms represent constant parameters (intercepts) in the model equations. The constants (C (3), C (4), C (5), C (6)) are parameters in the conditional variance specification (EGARCH-type model). They don’t have the same “intercept” meaning as in the mean equation; instead, they determine how volatility evolves. VOLREXR = real exchange rate volatility.

The diagnostic test analysis in Table 5 confirms that the EGARCH model is both well-specified and effective in modeling VOLREXR. The results confirm that high-volatility periods tend to occur in clusters, an important feature for risk modeling, and the model demonstrates a strong overall fit, with a log-likelihood value of 85.28, indicating a satisfactory alignment between the estimated specification and the observed data. The Akaike Information Criterion ($\text{AIC} = -1.7236$) and Schwarz Information Criterion ($\text{SIC} = -1.5591$) values suggest that the model is both efficient and well-fitted without unnecessary complexity. Diagnostic checks by the Durbin–Watson

statistic of 1.822, being close to the ideal value of 2, indicate the absence of significant serial correlation in the residuals of the mean equation, supporting correct model specification. Meanwhile, R^2 for the mean equation is very low (0.025).

The residuals remained from the estimated EGARCH model. The ARCH test, a key diagnostic tool proposed by Engle (1982), was conducted to determine if any time-varying volatility, or “volatility clustering”, The tests assess whether conditional heteroskedasticity persists after modeling; results show no evidence of ARCH effects, indicating the variance dynamics have been adequately captured. The presence of ARCH in Table 6, exploring the ARCH test effects, was assessed using the Lagrange Multiplier (LM) test. The computed ARCH F -statistic (0.0243) and its p -value (0.8764) indicate a failure to reject the null hypothesis of homoscedastic residuals, which was tested against the alternative of ARCH effects, providing no evidence of ARCH effects. Similarly, the ARCH ($\text{Obs} \times R^2$) statistic (0.0249) with a p -value of 0.8747 confirmed the absence of conditional heteroskedasticity. These results suggest that the residuals are well-behaved, validating the variance specification of the estimated model. To further evaluate model adequacy, the Durbin–Watson statistic was applied to detect serial correlation in the squared residuals being close to the benchmark of 2.0, revealing no significant autocorrelation, thereby reinforcing the robustness of the specification.

Table 5. Exponential generalized Autoregressive Conditional Heteroskedasticity Model diagnostic test results

Statistic	Value	Statistic	Value
R^2	0.025139	Mean dependent var	0.027136
Adjusted R^2	0.014307	S.D. dependent var	0.166848
S.E. of regression	0.165651	Akaike info criterion	-1.723585
Sum squared resid	2.469609	Schwarz criterion	-1.559120
Log likelihood	85.28489	Hannan-Quinn criter.	-1.657205
Durbin-Watson stat	1.822289		

Table 6. Heteroskedasticity test: Autoregressive Conditional Heteroskedasticity

Test statistic	Value	<i>p</i> -Value		
<i>F</i> -statistic (1,89)	0.024322	0.8764		
Obs × <i>R</i> ² χ^2 (1)	0.024862	0.8747		
Variable	Coefficient	Std. Error	<i>t</i> -Statistic	<i>p</i> -Value
C	1.024630	0.286288	3.579013	0.0006
WGT RESID ² (-1)	0.016524	0.105951	0.155956	0.8764
Statistic	Value	Statistic	Value	
<i>R</i> ²	0.000273	Mean dependent var	1.041808	
Adjusted <i>R</i> ²	-0.010960	S.D. dependent var	2.507105	
S.E. of regression	2.520806	Akaike info criterion	4.708768	
Sum squared resid	565.5473	Schwarz criterion	4.763951	
Log likelihood	-212.2489	Hannan-Quinn criterion	4.731031	
<i>F</i> -statistic	0.024322	Durbin-Watson stat	1.997625	
Prob (<i>F</i> -statistic)	0.876421			

2.10 Generating the Volatility Conditional Variance Series

The analysis supports evidence that the EGARCH model is well-specified. Due to the absence of significant autocorrelation in the residuals, this confirmed that the model has successfully captured the dynamic structure to generate the VOLREXR series, the independent variable in this research.

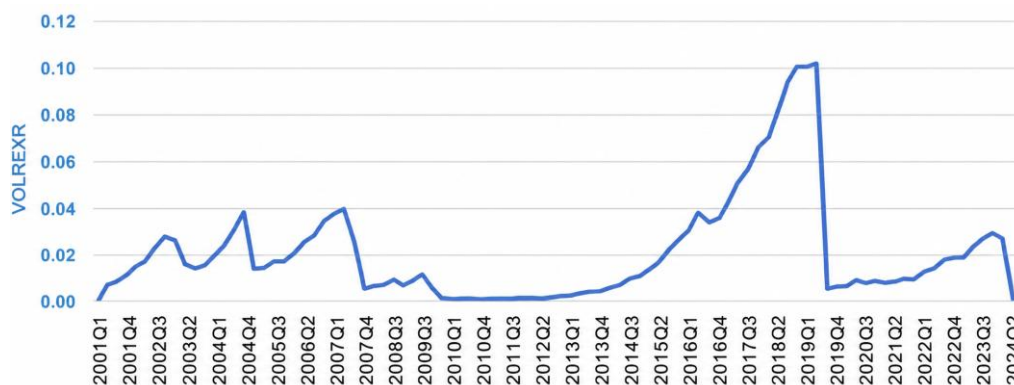


Figure 3. Real exchange rate volatility (VOLREXR) in Egypt 2001Q1–2024Q2

This finding, combined with the earlier ARCH test results (which indicated no remaining heteroscedasticity), consequently, the EGARCH framework, validates the model's empirical adequacy and its suitability for forecasting and subsequent econometric analysis of VOLREXR (Bollerslev, 1986; Engle, 1982). In this research, a new time-series variable was developed to capture the volatility of Egypt's real exchange rate. Figure 3 presents the VOLREXR series, generated in EViews 12, which is later used to assess the impact of VOLREXR on Egypt's aggregate FDI inflows, 2001–2024.

3. Results of Aggregate Foreign Direct Investment Econometric Autoregressive Distributed Lag Co-Integration Analysis

3.1 Unit Root (Augmented Dickey–Fuller and Phillips–Perron) Testing the Stationarity Results

To assess the time-series properties of the variables, the analysis employed the ADF and PP unit root tests, which are essential for verifying stationarity before conducting robust econometric estimations. The similarity between ADF and PP test results reinforces the reliability of these stationarity conclusions, Lajevardi & Chowdhury (2024). Applying conventional significance levels of 1%, 5%, and 10%, the results from both tests uniformly reject the null hypothesis of a unit root for aggregate FDI and the associated control variables. This outcome suggests that these series are stationary after first difference, indicating integration of order one, $I(1)$, as presented in the empirical results in Table 7.

Table 7. Augmented Dickey–Fuller (ADF) and Phillips–Perron (PP) unit root test, stationary results

Variables	Description	Test Statistic	<i>p</i> -Value	Stationary Status
LFDI	Aggregate foreign direct investment	-11.98781	<0.001	Stationary at $I(1)$.
VOLREXR	Exchange rate volatility	-9.000846	<0.001	Stationary at $I(1)$.
LCPI	Proxy of inflation	-8.950739	<0.001	Stationary at $I(1)$.
LGDP	Proxy of market size	-11.00335	<0.001	Stationary at $I(1)$.
LMCAP	Market capitalization	-8.714731	<0.001	Stationary at $I(1)$.
LEXDBTG	External debit	-9.206613	<0.001	Stationary at $I(1)$.

Note: Results are significant at 1%, 5 %, and 10 % levels, respectively.

3.2 The Optimal Lag Length Selection

Optimal lag selection is critical for cointegration analysis. AIC and Likelihood Ratio (LR) criteria suggest extending lags to 2–3 periods, better capturing interdependencies between aggregate/sectoral FDI and VOLREXR, compared to the initially specified ARDL (1, 0) model. These extended lags appear to better represent the short- and medium-run dynamics of the system. Consequently, these findings inform the specification of alternative ARDL models for subsequent estimation, as summarized in Table 8, based on AIC-driven model selection. Final Prediction Error (FPE) is lowest at lag 2, AIC is lowest at lag 3, and Schwarz criterion (SC), also known as the Bayesian information criterion (BIC) is lowest at lag 1. Hannan-Quinn (HQ) Criterion is lowest at lag 2, and the LR test favors lag 3. In the VAR lag order selection criteria table, the abbreviation NA stands for “Not Available”. Here's why:

- In the EViews output, the LR statistic is not reported for lag 0 because there is no previous lag to compare against.
- Therefore, the software displays NA to indicate that the statistic cannot be computed at that lag.
- For lags 1 and above, LR values are reported normally.

Table 8. Vector autoregression (VAR) lag order selection criteria

Lag	LogL	LR	FPE	AIC	SC	HQ
0	145.6238	NA	9.25e-10	-3.773615	-3.586799	-3.699092
1	632.1082	880.9312	4.79e-15	-15.94887	-14.64116*	-15.42721
2	695.8932	105.1591	2.30e-15*	-16.69982	-14.27121	-15.73101*
3	732.142653	88431*	2.40e-15	-16.70656*	-13.15705	-15.29062
4	765.7761	44.54162	2.81e-15	-16.64260	-11.97220	-14.77952
5	797.2884	36.62237	3.74e-15	-16.52131	-10.73001	-14.21109
6	832.4483	35.15993	4.98e-15	-16.49860	-9.586408	-13.74124
7	858.0692	21.46618	9.87e-15	-16.21809	-8.184996	-13.01359

Note: LR = Likelihood Ratio; FPE = Final Prediction Error; AIC = Akaike Information Criterion; SC = Schwarz Criterion; HQ = Hannan-Quinn; NA = Not Available. * = Optimal lag length selection.

3.3 The Short Run Dynamic Autoregressive Distributed Lag of Real Exchange Rate Volatility on Foreign Direct Investment

The short-run estimates from the ARDL (1, 0) specification, as reported in Table 9, indicate a statistically significant and negative relationship between VOLREXR and aggregate FDI in Egypt. Specifically, the coefficient of -11.52 ($p = 0.009$) suggests that, holding other factors constant, a one-unit rise in VOLREXR is associated with an immediate reduction of approximately 11.52 units in aggregate FDI inflows during the 2001–2024 period. Furthermore, in Table 9, the lagged dependent variable, LFDI (-1), exhibits a positive and highly significant coefficient of 0.4938 ($t = 5.7165$, $p < 0.001$), implying that nearly 49.4% of the previous quarter's FDI continues in the current period, reflecting a high degree of persistence in FDI flows, highlighting the role of historical FDI in shaping current inflows. These findings confirm a robust short-run association between VOLREXR and FDI, validating the ARDL framework as an appropriate econometric approach for modeling the dynamic interactions between variables.

Table 9. The short-run Autoregressive Distributed Lag Model co-integration of aggregate (FDI)

Variable	Coefficient	Std. Error	t-Statistic	p-Value *
LFDI (-1)	0.493793	0.086380	5.716502	0.0000
VOLREXR	-11.51935	4.304712	-2.675986	0.0090
LGDP	0.569726	0.141814	4.017412	0.0001
LCPI	1.288605	0.352939	3.651073	0.0005
LMCAP	1.042517	0.237238	4.394399	0.0000
LEXDBTG	5.729818	1.818813	3.150305	0.0023
C	-16.43180	4.121816	-3.986543	0.0001
R^2	0.792816	Mean dependent var		7.194481
Adjusted R^2	0.777469	S.D. dependent var		1.264922
S.E. of regression	0.596704	Akaike info criterion		1.881413
Sum squared resid	28.84052	Schwarz criterion		2.078474
Log likelihood	-75.78216	Hannan-Quinn criter.		1.960804
F-statistic	51.65945	Durbin-Watson stat		2.023522
Prob (F-statistic)	0.000000			

Note: FDI = foreign direct investment; LFDI = aggregate foreign direct investment; VOLREXR = real exchange rate volatility; LGDP = proxy of market size; LCPI = proxy of inflation; LMCAP = market capitalization; LEXDBTG = External debit; C = constant term (intercept).

Within the ARDL framework presented in Table 10 and Table 11, the estimated error correction term, ECT (-1), is negative and highly significant ($p < 0.001$), satisfying key diagnostic requirements and confirming the existence of a stable long-run cointegrating relationship between FDI, VOLREXR, and other control variables GDP, CPI, MCAP, and EXDBTG. The ECT (-1) coefficient of -0.5062 indicates that approximately 50.6% of any disequilibrium from the preceding period is corrected within the current period, reflecting a relatively rapid adjustment toward the long-run equilibrium. This finding highlights the strong tendency of FDI to revert to its equilibrium path in response to deviations driven by VOLREXR. In contrast, the short-run coefficients capture the immediate directional effects of changes in (VOLREXR) on FDI flows, reflecting the distinction between temporary impacts and the long-term adjustment process.

The negative impact of the exchange rate volatility in this study aligns with classical economic theory, which maintains that heightened exchange rate uncertainty increases investment risk and diminishes the predictability of returns, thereby deterring foreign capital inflows. This outcome aligns with prior empirical evidence from Egypt, including Asamoah et al. (2022), Elagouza (2023), Elbadry & Mandour (2020), Elmoghlar (2025), and Marwa et al. (2024), all of which emphasize the importance of currency stability in attracting FDI, particularly relevant during Egypt's transition toward a more flexible exchange rate regime. Comparable findings have been reported in international contexts, such as Abdel Aziz & Al-Ajrawi (2023), Akinlo & Onatunji (2020), Akinlo & Onatunji (2021), El Rhadbane & El Moudden (2022), Faizi et al. (2024), Fatima et al. (2023), Hamida (2024), Hniya et al. (2021), Imamboccus et al. (2024), Jamal & Bhat (2022), Jannat (2020), Ozigbo & Anuya (2023), Qamruzzaman et al. (2021), Rasheed et al. (2020), Udoinyang & Udoinyang (2024), and Zongo et al. (2024).

Table 10. Autoregressive Distributed Lag error correction model regression dependent variable

Variable	Coefficient	Std. Error	t-Statistic	p-Value
LGDP	0.569726	0.134722	4.228896	0.0001
LCPI	1.288605	0.292873	4.399877	0.0000
LMCAP	1.042517	0.182061	5.726182	0.0000
LEXDBTG	5.729818	1.184015	4.839313	0.0000
CointEq (-1)	-0.506207	0.080501	-6.288202	0.0000

Note: LGDP = proxy of market size; LCPI = proxy of inflation; LMCAP = market capitalization; LEXDBTG = External debt.

Table 11. Short-run dynamics in the Autoregressive Distributed Lag Model for net FDI

Variables	Coefficient	Std. Error	t-Statistic	p-Value
ECT (-1) *	-0.506207	0.086380	-5.860221*	0.0000
LFDI (-1)	0.493793	0.086380	5.716502*	0.0000
VOLREXR*	-11.51935	4.304712	-2.675986*	0.0090
LGDP	0.569726	0.141814	4.017412*	0.0001
LCPI	1.288605	0.352939	3.651073*	0.0005
LMCAP	1.042517	0.237238	4.394399*	0.0000
LEXDBTG	5.729818	1.818813	3.150305*	0.0023
C	-16.43180	4.121816	-3.986543*	0.0001
R^2 : 0.792816		Adjusted R^2 : 0.777469		

Note: FDI = foreign direct investment; ECT = error-correction term; LFDI = aggregate foreign direct investment; VOLREXR = real exchange rate volatility; LGDP = proxy of market size; LCPI = proxy of inflation; LMCAP = market capitalization; LEXDBTG = External debit; C = constant term (intercept). * Significant at 1%, 5%, and 10% levels, respectively.

Conversely, some strands of the literature result in a positive impact of the VOLREXR on FDI, frequently attributing this outcome to the investment opportunities that arise during periods of currency fluctuation. Evidence supporting this perspective within the Egyptian context has also been reported by Abdelgany (2020), Al-Ghiash (2022), Al-Sayed (2020), Fadl & Ghoneim (2020), Hend et al. (2021), Nadine et al. (2021). At the international level, comparable findings have been reported, indicating that similar patterns are observed across diverse economies like Ejaz & Azam (2024), Ezekiel & Temidayo (2023), Huong et al. (2020), Kaya & Erden (2022), Lajevardi & Chowdhury (2024), Lin & Chen (2022), Okonkwo et al. (2021), Shafique et al. (2022), Sultana et al. (2024), Thujijanthan (2021), Zaharum et al. (2024).

3.4 Co-Integration Analysis Using Autoregressive Distributed Lag F-Bounds Test

To address potential issues of model misspecification and serial correlation, this study employs the ARDL bound testing methodology. Model specification was guided by the AIC, which identified an optimal lag structure of one period for both dependent and independent variables. The F-bounds test was applied to evaluate the null hypothesis of no long-run equilibrium impact of the VOLREXR on FDI alongside key macroeconomic control variables. As reported in Table 12 and Table 13, the computed F-statistic (12.8629) substantially exceeds the upper-bound critical values at all conventional significance levels. Otherwise, exceeding the 1% upper-bound thresholds for both asymptotic (5.58) and finite-sample (5.917) distributions, thereby leading to rejection of the null hypothesis at the 1% significance level. These results provide robust empirical evidence of a stable long-run cointegrating relationship between FDI and VOLREXR, a relationship that remains significant even after controlling for market size (GDP), inflation (CPI), financial MCAP, and EXDBTG.

Table 12. Upper and lower bound test statistics

Significance Level	Asymptotic Critical Values ($n = 1000$)		Finite Sample Critical Values ($n = 80$, Approximation for $n = 88$)	
	Lower Bound (I(0))	Upper Bound (I(1))	Lower Bound (I(0))	Upper Bound (I(1))
10%	3.02	3.51	3.113	3.61
5%	3.62	4.16	3.74	4.303
2.5%	4.18	4.79	-	-
1%	4.94	5.58	5.157	5.917
Compare the upper and lower bounds with the test statistic F -statistic: 12.86289				

Note: F-Bounds test: F -statistics (12.86289) is greater than the upper bound critical values at all conventional significance levels (10%, 5%, 2.5%, 1%) for both I(0) and I(1) variables; "-" indicates not applicable.

Table 13. F-bounds test for co-integration

Variable	Conditional Error Correction Coefficient	Calculation (Coefficient/-(-0.506207))	Long-Run Coefficient	p-Value (Significance Level)
C	-16.43180	-16.43180/0.506207	32.46062	0.0001*
VOLREXR	-11.51935	-11.51935/0.506207	-22.75619	0.0090*
LGDP	0.569726	0.569726/0.506207	1.12544	0.0001*
LCPI	1.288605	1.288605/0.506207	2.54561	0.0005*
LMCAP	1.042517	1.042517/0.506207	2.05942	0.0000*
LEXDBTG	5.729818	5.729818/0.506207	11.31911	0.0023*

Note: C = constant term (intercept); VOLREXR = real exchange rate volatility; LGDP = proxy of market size; LCPI = proxy of inflation; LMCAP = market capitalization; LEXDBTG = external debit. Significance Level: All long-run coefficients derived are statistically

significant at commonly accepted levels (e.g., 1% or 5%), as indicated by their corresponding p -values. The long-run coefficients are obtained by dividing the coefficients of the independent variables by the negative of the coefficient of $LFDI(-1)$ *. Coefficient of $LFDI(-1)$ * = -0.506207.

3.5 The Long Run Dynamic Autoregressive Distributed Lag of Real Exchange Rate Volatility on Aggregate Foreign Direct Investment

The study examines the long-run dynamics of FDI in Egypt using a Conditional Error Correction Model (CECM). The long-run coefficients were estimated using VOLREXR as the primary explanatory variable, with the adjustment speed captured by the negative coefficient of the lagged dependent variable, $LFDI(-1)$ (-0.506207). Empirical results, as presented in Table 14, in the long run reveal a statistically significant and negative impact of VOLREXR on aggregate FDI inflows. Specifically, the estimated coefficient of -22.75619 ($t = -2.676$, $p = 0.009$) indicates that a 1% rise in VOLREXR corresponds to an approximate 22.76% decline in FDI over the long run (see Table 14).

Table 14. Autoregressive Distributed Lag long run form and Bounds test

Variable	Coefficient	Std. Error	t-Statistic	p-Value
VOLREXR	-22.75619	8.962496	-2.539047	0.0130
C	-32.46062	6.682087	-4.857856	0.0000
ECT = $LFDI - (-22.7562 \times VOLREXR - 32.4606)$				

Note: VOLREXR = real exchange rate volatility; C = constant term (intercept); ECT = error-correction term; LFDI = aggregate foreign direct investment.

The results align with theoretical perspective expectations and corroborate prior empirical findings for Egypt, as documented by Asamoah et al. (2022), Elagouza (2023), Elbadry & Mandour (2020), Elmoghriar (2025), and Marwa et al. (2024). Over the period 2001Q1–2024Q2, sustained VOLREXR and recurrent macroeconomic instability appear to have hindered the attraction of aggregate FDI inflows. This negative impact is reasonably attributable to heightened transaction and currency conversion costs, which increase operational uncertainty and reduce the long-term appeal of investment in the Egyptian market.

Overall, the empirical evidence supports the hypothesis that VOLREXR exerts a significant and negative influence on Egypt's aggregate FDI inflows in the long run, offering important policy implications for promoting macroeconomic stability and strengthening the country's investment climate. Nevertheless, these findings stand in contrast to a subset of studies on the Egyptian economy, where several investigations report alternative outcomes for FDI, such as Abdelgany (2020), Al-Ghiash (2022), Al-Sayed (2020), Fadl & Ghoneim (2020), Hend et al. (2021), and Nadine et al. (2021). These studies reported a positive exchange rate volatility effect on FDI; in these cases, exchange rate depreciation is argued to enhance the competitiveness of local assets and production inputs, thereby attracting foreign investors seeking cost advantages.

The adverse relationship identified in this study is consistent with a substantial number of previous international studies, including contributions from scholars such as Abdel Aziz & Al-Ajrabi (2023), Akinlo & Onatunji (2020), Akinlo & Onatunji (2021), El Rhadbane & El Moudden (2022), Faizi et al. (2024), Fatima et al. (2023), He (2018), Hniya et al. (2021), Imamboccus et al. (2024), Jamal & Bhat (2022), Jokosenumi & Adesete (2018), Latief & Lefen (2018), Musyoka (2018), Ozigbo & Anuya (2023), Qamruzzaman et al. (2021), Rasheed et al. (2020), Udoinyang & Udoinyang (2024), and Zongo et al. (2024). Moreover, an economy's demonstrated capacity to absorb and adapt to exchange rate fluctuations can serve as an indicator of structural resilience, potentially shaping investor confidence and sustaining FDI inflows. This perspective is reinforced by evidence drawn from various international contexts, such as Ezekiel & Temidayo (2023), Kaya & Erden (2022), Lajevardi & Chowdhury (2024), Lin & Chen (2022), Okonkwo et al. (2021), and Shafique et al. (2022).

3.6 Control Variables Effect in The Short-Long Run on Foreign Direct Investment in Egypt

A. Market Size Effect in the Short and Long Run:

The ARDL estimation results demonstrate that Egypt's market size during 2001–2024, measured by GDP, is a significant determinant of FDI inflows in both the short and long run. In the short run, the positive and statistically significant coefficient (0.5697; $t = 4.017$, $p < 0.01$) suggests that a one-unit increase in GDP corresponds to an approximate 0.57 unit rise in logged FDI, holding other factors constant. This supports the market-seeking FDI hypothesis, indicating that economic expansion strengthens investor confidence by signaling higher domestic demand, greater sales potential, and improved profitability prospects (Elmoghriar, 2025; Hamida, 2024; Lajevardi & Chowdhury, 2024;). Meanwhile, the results in the long-run estimates reinforce this conclusion, with a coefficient of 1.1254 ($t = 4.017$, $p = 0.0001$) confirming that sustained economic growth is a core driver of FDI inflows. A larger and expanding market offers foreign investors access to a broader consumer base and opportunities for economies of scale. These findings are consistent with the broader empirical literature on FDI determinants (Arvin

et al., 2021) and are strongly supported by both Egyptian-specific evidence Elmoghiar (2025) and Nadine et al. (2021). Overall, the results highlight the critical role of macroeconomic stability and growth in attracting market-oriented investment to emerging economies such as Egypt.

B. Consumer Price Index Impact in the Short and Long Run:

The ARDL analysis highlights a significant role for inflation, proxied by the CPI, in shaping FDI inflows in Egypt across the study period (2001–2024). In the short run, the positive and significant coefficient (1.2886; $t = 3.65$, $p < 0.01$) suggests that higher inflation rates were associated with increased FDI inflows. In the long run, ARDL estimates reinforce the positive association, with a coefficient of 2.5456 ($t = 3.651$, $p = 0.0005$), indicating that continuous inflation was associated with FDI growth. The positive effect of the CPI on FDI observed in this study warrants further interpretation. One possible explanation is that moderate increases in CPI may reflect rising domestic demand and expanding market size, which can be perceived by foreign investors as an indicator of growth potential. Additionally, inflationary pressures may reduce the real cost of domestic production inputs in foreign currency terms, thereby enhancing competitiveness and attracting investment. In this sense, CPI may serve as a proxy for dynamic economic activity rather than purely as a measure of instability. In Egypt's case, the result may reflect the CBE's consistent inflation-targeting framework, where moderate and stable inflation signaled policy credibility, thereby mitigating perceived investment risk. This finding contrasts with empirical literature, as studies such as Imamboccus et al. (2024) reported that inflation discourages FDI. Provided in Egypt, during 2001 to 2024, an increase in GDP stimulates higher demand for goods and services, which raises domestic prices and reflects short-run inflationary tendencies, which are interpreted positively as a growing market with profitable FDI opportunities, entering to capture share despite rising costs. In addition, GDP growth is linked to improved infrastructure and stronger macroeconomic fundamentals, which reduce investment risk and enhance the Egyptian economy's attractiveness. Thus, despite inflationary pressures, overall growth encourages greater FDI inflows, as investors seek long-term returns in a dynamic, expanding market.

C. Market Capitalization:

In the short run, the empirical results show a significant positive relationship between MCAP and FDI net inflows. In the short run, the estimated coefficient (1.0425; $t = 4.394$, $p < 0.01$) indicates that rising equity market valuations substantially enhance foreign investment. Meanwhile, long-run estimates further strengthen this conclusion, with a coefficient of 2.0594 ($t = 4.394$, $p < 0.01$) suggesting that sustained growth in MCAP is linked to persistent FDI inflows. This supports theoretical perspectives that emphasize the role of deep, liquid capital markets in attracting international capital by offering improved access to financing, portfolio diversification opportunities, and efficient exit mechanisms. Reforms implemented by the Egyptian exchange to strengthen secondary markets have likely reinforced this effect by increasing asset allocation flexibility and lowering transaction costs for foreign investors. A mature capital market signals macroeconomic resilience, strong corporate performance, and policy stability, factors that encourage long-run foreign investment. These findings highlight a mutually reinforcing relationship between equity market development and FDI.

D. External Debt:

This study's findings align with prior evidence suggesting that elevated debt burdens can heighten investor risk perceptions and reduce capital inflows in emerging markets. In the short run, the EXDBTG growth exhibits a negative and statistically significant impact ($\beta = -5.71$, $p = 0.048$), implying that increased borrowing may initially deter foreign investors by heightening concerns over country risk, repayment capacity, and macroeconomic stability (El Mahdy et al., 2022). Conversely, the long-run results reveal a positive and significant association ($\beta = 11.32$, $t = 3.15$, $p = 0.002$), indicating that EXDBTG can support sustained FDI inflows when strategically managed. This outcome is consistent with theoretical perspectives that emphasize the role of debt-financed resources in funding infrastructure, energy transition, and large-scale development projects that enhance the investment climate. A notable example occurred in 2024, when the CBE reported a USD 11.8 billion (7.2%) reduction in EXDBTG, largely due to converting approximately USD 11 billion in UAE deposits into equity for the Ras El Hekma development project. This transaction illustrates how debt restructuring can be leveraged to attract FDI by transforming liabilities into productive investment. More broadly, reasonable borrowing can reinforce macroeconomic stability, stimulate domestic savings, and crowd in private capital flows. The positive effect of EXDBTG on FDI observed here may therefore reflect investor confidence in debt-financed development strategies, where external resources are deployed to expand productive capacity, improve competitiveness, and signal long-term growth potential.

3.7 Diagnostic Autoregressive Distributed Lag Test Statistics for Aggregate Foreign Direct Investment

Following the methodological frameworks outlined in the literature by Abdel Aziz & Al-Ajrawi (2023), Elmoghiar (2025), and Lajevardi & Chowdhury (2024), this research categorizes its diagnostic procedures to investigate the impact of VOLREXR on aggregate FDI. The econometric evaluation employed the Durbin–Watson d-statistic and the Breusch–Pagan–Godfrey test to detect serial correlation in model residuals and the VIF to test multicollinearity. Heteroscedasticity was assessed using the White test, while the Jarque–Bera statistic was applied

to examine the normality of residual distributions. Furthermore, the structural stability of ARDL model parameters was evaluated through the CUSUM and CUSUM of squares tests. Collectively, these diagnostic measures were implemented to ensure the robustness, validity, and reliability of empirical findings.

3.8 The Autoregressive Distributed Lag Model Diagnostic Test and Goodness of Fit

Consistent with methodological practices in prior empirical research, the ARDL analysis presented in Table 15 demonstrates strong model adequacy in explaining aggregate FDI inflows in Egypt. The Durbin–Watson statistic ($d=2.0235$) is close to the benchmark value of 2, suggesting the absence of significant autocorrelation in the residuals. The coefficient of determination ($R^2=0.7928$) indicates that VOLREXR, together with key control variables, explains approximately 79% of the variation in FDI, reflecting a high degree of explanatory power that may also influence FDI, thereby ensuring a more comprehensive and reliable model specification. Economically, the negative relationship suggests that heightened volatility increases uncertainty and risk for foreign investors, discouraging long-term commitments and reducing capital inflows. This aligns with theories of investment under uncertainty, where exchange rate instability raises transaction costs, complicates profit repatriation, and undermines confidence in the host economy.

Table 15. Autoregressive Distributed Lag Model co-integration result

Statistic	Value	Statistic	Value
R^2	0.792816	Mean dependent var	7.194481
Adjusted R^2	0.777469	S.D. dependent var	1.264922
S.E. of regression	0.596704	Akaike info criterion	1.881413
Sum squared resid	28.84052	Schwarz criterion	2.078474
Log likelihood	-75.78216	Hannan-Quinn criterion	1.960804
F -statistic	51.65945	Durbin-Watson stat	2.023522
Prob (F -statistic)	0.000000		

Note: S.E. = standard error; S.D. = standard deviation.

The adjusted R^2 (0.7775) further confirms the robustness of this fit after accounting for the number of predictors. The F -statistic (51.659, $p < 0.001$) demonstrates that VOLREXR, GDP, CPI, MCAP, and EXDBTG exert a statistically significant joint influence on FDI inflows. Model selection metrics, including the Akaike Information Criterion (AIC=1.8814) and Schwarz Criterion (SC=2.0785), provide additional support for the model's suitability. Nevertheless, while the model explains a relatively high proportion of variation, the possibility of omitted variables should be acknowledged. Factors such as political stability, institutional quality, trade openness, and global financial conditions may also influence FDI inflows but are not explicitly captured in the present specification. Their exclusion could introduce bias or limit the generalizability of results. Thus, while the model demonstrates strong fit and statistical adequacy, future research may benefit from incorporating these broader determinants to provide a more comprehensive understanding of FDI dynamics in Egypt.

3.9 Autocorrelation Test (Breusch–Pagan–Godfrey)

In line with the methodology proposed by Breusch (1978) and Godfrey (1978), the Breusch–Godfrey Serial Correlation LM test was applied to evaluate whether the regression residuals were independent, thereby ensuring the reliability of parameter estimates. This diagnostic is widely recognized in econometric analysis for verifying the assumption of independently distributed errors, which is essential for producing unbiased coefficients and valid statistical inference. For the ARDL model examining the impact of VOLREXR and FDI in Egypt, the test results for lag orders one and two yielded p -values of 0.5005 and 0.4657, respectively, both exceeding the 0.05 significance level. Consequently, the null hypothesis of no serial correlation could not be rejected. This finding in Table 16 indicates that residuals are free from statistically significant autocorrelation up to two lags, thereby fulfilling a fundamental regression assumption. The absence of autocorrelation enhances the credibility of the parameter estimates and reinforces the robustness of the ARDL model's conclusions regarding the impact of VOLREXR on FDI in Egypt over the period 2001–2024.

Table 16. Serial correlation test. Breusch–Godfrey Serial Correlation Lagrange Multiplier (LM) test results

Test Statistic	Value	p -Value (Distribution)	Value
F -statistic	0.698258	p -value F (2, 79)	0.5005
Obs $\times R^2$	1.528592	p -value Chi-Square (2)	0.4657

3.10 Multicollinearity Test Analysis of Variance Inflation Factor Results

The research employed the VIF method as a diagnostic tool for multicollinearity to ensure the reliability of the data of the estimated impact of VOLREXR on FDI in Egypt, VIF can detect the independent variables that are highly correlated to each other because the multicollinear variables can affect the robustness of the estimated model by generating biased coefficients.

The results in Table 17 indicate that all centered VIF values fall below the conventional threshold of 10, suggesting that multicollinearity is not a serious concern in the model. Specifically, the lagged dependent variable (LFDI (-1)) recorded a VIF of 2.86, while the exchange rate volatility variable (VOLREXR) showed a value of 2.52. Other control variables, including GDP (1.89), consumer prices CPI (2.37), MCAP (3.14), and EXDBTG (2.71), also remained within acceptable ranges. These findings confirm that the explanatory variables are sufficiently independent, thereby ensuring that the estimated coefficients are not inflated by hidden correlations. This is particularly important in finance and macroeconomic models, where variables such as exchange rates, inflation, and EXDBTG often exhibit interdependence (Salmerón-Gómez et al., 2025). By incorporating VIF alongside other diagnostic tests, including Durbin–Watson for autocorrelation, the study provides a comprehensive robustness framework. Thus, the inclusion of VIF strengthens the validity of the econometric results by confirming that the observed effects of VOLREXR on aggregate FDI flow in Egypt, 2001–2024, are not distortions of multicollinearity but reflect valid economic relationships.

Table 17. Variance Inflation Factors (VIF)

Variable	Coefficient Variance	Uncentered VIF	Centered VIF
LFDI (-1)	0.007462	96.37530	2.858288
VOLREXR	18.53054	4.495327	2.520305
LGDP	0.020111	85.18579	1.894349
LCPI	0.124566	7.417539	2.374715
LMCAP	0.056282	1,629.723	3.136444
LEXDBTG	3.308082	966.7996	2.707930
C	16.98937	4,198.960	NA

Note: LFDI = aggregate foreign direct investment; VOLREXR = real exchange rate volatility; LGDP = proxy of market size; LCPI = proxy of inflation; LMCAP = market capitalization; LEXDBTG = external debt; C = constant term (intercept); NA = not available.

3.11 Heteroskedasticity Test (Breusch-Pagan-Godfrey)

In accordance with standard econometric procedures, the Breusch–Pagan–Godfrey test was applied to determine whether the variance of the regression error term remained constant across observations. This diagnostic assesses whether residual variability is systematically related to the explanatory variable, the VOLREXR, a condition that, if present, could distort statistical inference. As presented in Table 18, the p -values for the F -statistic (0.0528) and the $\text{Obs} \cdot R^2$ statistic (0.0565) were slightly above the conventional 5% significance threshold. Consequently, the null hypothesis of homoskedasticity could not be rejected, indicating no statistically significant evidence of heteroskedasticity in the residuals. This finding affirms the reliability of the estimated standard errors and supports the validity of the empirical results derived from the ARDL model examining the impact of VOLREXR on FDI in Egypt, 2001–2024. Furthermore, all estimations incorporated heteroskedasticity-robust standard errors to mitigate any potential effects of variance misspecification.

Table 18. Heteroskedasticity test: Breusch-Pagan-Godfrey test results

Test Statistic	Value	p -Value (Distribution)	Value
F -statistic	2.184314	p -value F (6, 81)	0.0528
$\text{Obs} \cdot R^2$	12.25553	p -value chi-square (6)	0.0565
Scaled explained SS	13.84892	p -value chi-square (6)	0.0314

Note: SS = sum of squares.

3.12 Cumulative Sum Structural Stability Test

Following established econometric diagnostics procedures for assessing parameter constancy, the CUSUM test was employed to examine the stability of the estimated regression coefficients for Egypt over the period 2001–2024. This technique evaluates the CUSUM of recursive residuals to identify potential structural breaks, abrupt shifts in the relationships among variables, that could undermine model validity. Inspection of the CUSUM plot (Figure 4) showed that the CUSUM path (blue line) remained entirely within the 5% significance bounds (red lines) throughout the study period. This result provides statistical evidence of parameter stability, indicating that the ARDL model is structurally sound and free from detectable structural breaks at conventional significance levels.

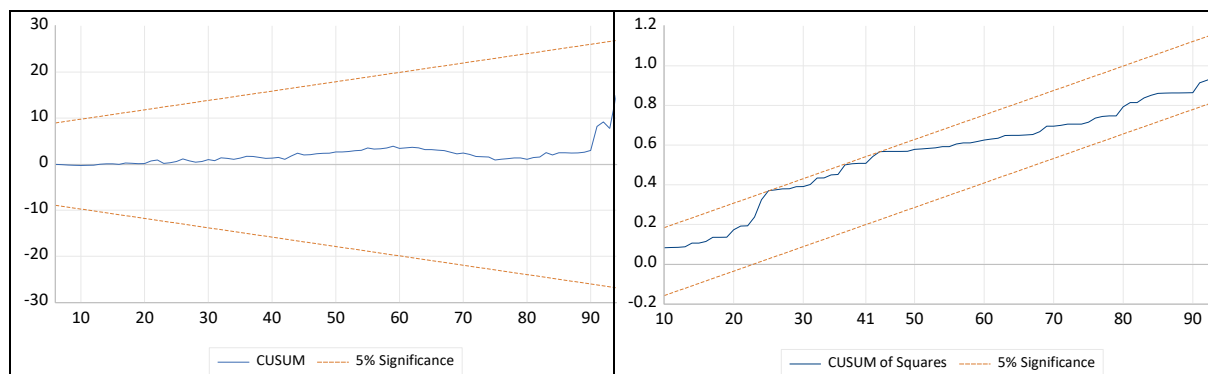


Figure 4. Cumulative Sum (CUSUM) of Squares (CUSUMQ)

Note: Horizontal axis (x-axis): Represents the time dimension (sample period), typically measured in observations (e.g., years, quarters, dataset). Each point corresponds to one observation in the sample. Vertical axis (y-axis): Shows the cumulative sum of squares of recursive residuals, which is used to test the stability of the regression model. Interpretation: If the CUSUMQ line remains within the 5% significance boundaries (usually shown as two straight lines), the model is considered stable over the sample period. Crossing the boundaries indicates potential structural instability.

3.13 Analysis of Residual Normality (Jarque-Bera Test)

Following established econometric diagnostic procedures, Jarque & Bera (1980) were applied to examine whether the residuals of the ARDL model were compatible with the assumption of normal distribution. This parametric test evaluates deviations from normality by jointly considering skewness and kurtosis in the data illustrated in Figure 5, both of which are critical for ensuring the validity of empirical statistics. The descriptive statistics indicated skewness values near zero and kurtosis close to the Gaussian benchmark of 3.3. The computed Jarque–Bera statistic was 1.683, with an associated p -value of 0.432, exceeding the standard 5% level of statistical significance. These findings suggest that the null hypothesis of residual normality cannot be rejected, implying that the ARDL model’s residuals are consistent with the normality assumption and that statistical constructs derived from the ARDL model are reliable.

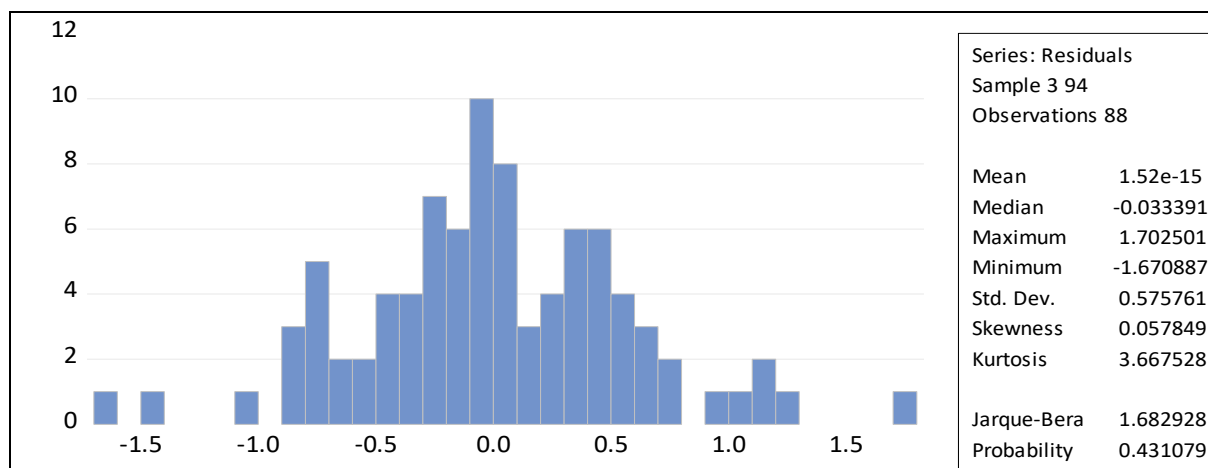


Figure 5. Jarque–Bera normality test

Note: Horizontal axis (x-axis): Represents the fitted values or residuals distribution bins. It shows the range of residuals from the regression model, grouped into intervals. Vertical axis (y-axis): Represents the frequency (count) of residuals falling into each bin.

4. Discussion of Results

The present study has investigated the impact of exchange rate volatility on FDI in Egypt, 2001–2024, with empirical evidence using the EGARCH-ARDL approach. The study formulates the exchange rate volatility using EGARCH. The diagnostic test analysis in Table 5 confirms that the EGARCH model is both well-specified and effective in modeling VOLREXR. The results confirm that high-volatility periods tend to occur in clusters, an important feature for risk modeling, and the model demonstrates a strong overall fit. The study answered the research questions and explored and understood how VOLREXR affects FDI composition, providing evidence that can inform the financial, macroeconomic policy, and investment strategies in Egypt, and also answered how VOLREXR impacts the inflows of aggregate FDI into Egypt in the short and long run during the period 2001–

2024. And also, the aggregate FDI empirical results and ARDL co-integration confirm the analysis hypothesis testing H1, demonstrating that VOLREXR has a statistically significant negative impact on aggregate FDI in both the short and long run in Egypt, offering critical policy implications for promoting macroeconomic stability and enhancing Egypt's investment climate.

However, the empirical ARDL confirmed a robust short-run and long-run cointegration between VOLREXR and FDI, validating the ARDL framework as an appropriate econometric approach for modeling the dynamic interactions between variables Ben-Obi et al. (2025). The analysis provides several important insights into the determinants of FDI inflows in Egypt. Consistent with classical economic theory, the study identifies a negative short-run and long-run relationship between VOLREXR and FDI, supporting the role of heightened uncertainty in elevating investment risk and reducing the predictability of returns. This finding aligns with a substantial body of international research that views exchange rate instability as a constraint to foreign capital inflows. These negative findings are aligned with prior empirical evidence on Egypt, reinforcing the view that exchange rate instability remains a critical barrier to sustaining foreign investment, including Asamoah et al. (2022), Elagouza (2023), Elbadry & Mandour (2020), Elmoghlar (2025), and Marwa et al. (2024), all of which emphasize the importance of exchange rate stability in attracting FDI, particularly relevant during Egypt's transition toward a more flexible exchange rate regime. Comparable findings have been reported in international contexts, such as Abdel Aziz & Al-Ajrawi (2023), Akinlo & Onatunji (2020), Akinlo & Onatunji (2021), El Rhadbane & El Moudden (2022), Faizi et al. (2024), Fatima et al. (2023), Hamida (2024), Hniya et al. (2021), Imamboccus et al. (2024), Jamal & Bhat (2022), Jannat (2020), Ozigbo & Anuya (2023), Qamruzzaman et al. (2021), Rasheed et al. (2020), Udoinyang & Udoinyang (2024), and Zongo et al. (2024). In contrast, other literature reports a positive relationship between VOLREXR and FDI, often attributing it to increased investment opportunities during periods of currency fluctuation. Within Egypt, such evidence is provided by Abdelgany (2020), Al-Ghiash (2022), Al-Sayed (2020), Fadl & Ghoneim (2020), Hend et al. (2021), and Nadine et al. (2021). Internationally, similar positive results are reported by Ejaz & Azam (2024), Ezekiel & Temidayo (2023), Huong et al. (2020), Kaya & Erden (2022), Lajevardi & Chowdhury (2024), Lin & Chen (2022), Okonkwo et al. (2021), Shafique et al. (2022), Sultana et al. (2024), Thujiyanthan (2021), and Zaharum et al. (2024). Moreover, the ARDL model demonstrates strong explanatory power and statistical adequacy. The Durbin–Watson statistic ($d = 2.0235$) suggests the absence of significant autocorrelation, while the coefficient of determination ($R^2 = 0.7928$) indicates that exchange rate volatility and the included control variables explain nearly 79% of the variation in FDI inflows. The adjusted R^2 (0.7775) confirms robustness after accounting for predictors, and the F -statistic (51.659, $p < 0.001$) highlights the joint significance of VOLREXR, GDP, CPI, MCAP, and EXDBTG. Model selection criteria (AIC = 1.8814; SC = 2.0785) further validate the suitability of the specification. Nevertheless, the possibility of omitted variables such as political stability, institutional quality, trade openness, and global financial conditions should be acknowledged, as these factors may also influence FDI inflows. Overall, the empirical evidence supports the hypothesis that VOLREXR exerts a significant and negative influence on Egypt's aggregate FDI inflows in the short run and long run, offering important policy implications for promoting macroeconomic stability and strengthening the country's investment climate.

Furthermore, the control variables analysis found that a larger and expanding market offers foreign investors access to a broader consumer base and opportunities for economies of scale. These findings are consistent with the broader empirical literature on FDI determinants (Arvin et al., 2021) and are strongly supported by both Egyptian-specific evidence (Elmoghlar, 2025; Nadine et al., 2021), highlighting the critical role of macroeconomic stability and growth in attracting market-oriented investment to emerging economies such as Egypt. The ARDL analysis highlights a significant role for inflation, proxied by the CPI, in shaping FDI inflows in Egypt across the study period (2001–2024). One possible explanation is that moderate increases in CPI may reflect rising domestic demand and expanding market size, which can be perceived by foreign investors as an indicator of growth potential. Additionally, inflationary pressures may reduce the real cost of domestic production inputs in foreign currency terms, thereby enhancing competitiveness and attracting investment. The MCAP and FDI net inflows empirical results show a significant positive relationship in the short run and long run, indicating that rising equity market valuations substantially enhance foreign investment, suggesting that sustained growth in MCAP is linked to persistent FDI inflows. This supports theoretical perspectives that emphasize the role of deep, liquid capital markets in attracting international capital by offering improved access to financing, portfolio diversification opportunities, and efficient exit mechanisms. In addition, the EXDBTG burdens can heighten investor risk perceptions and reduce capital inflows in emerging markets. In the short run, the EXDBTG growth exhibits a negative and statistically significant impact, implying that increased borrowing may initially deter foreign investors by heightening concerns over country risk, repayment capacity, and macroeconomic stability (El Mahdy et al., 2022). Conversely, the long-run results reveal a positive and significant result, indicating that EXDBTG can support sustained FDI inflows when strategically managed. This outcome is consistent with theoretical perspectives that emphasize the role of debt-financed resources in funding infrastructure, energy transition, and large-scale development projects that enhance the investment climate.

Following the methodological frameworks established in recent scholarship by Abdel Aziz & Al-Ajrawi (2023),

Lajevardi & Chowdhury (2024), and Elmoghriar (2025), the ARDL model employed in this study demonstrates strong statistical adequacy and a reliable overall fit. The diagnostic procedures reinforce the robustness of empirical findings. Specifically, the Durbin–Watson statistic and the Breusch–Pagan–Godfrey test confirms the absence of autocorrelation in the residuals, thereby enhancing the credibility of the parameter estimates. Similarly, the VIF indicates no evidence of multicollinearity, ensuring that the observed negative relationship between exchange rate volatility and FDI inflows reflects credible economic dynamics rather than statistical distortions. The White test further suggests no significant heteroscedasticity, affirming the reliability of the estimated standard errors, while the Jarque–Bera statistic confirms that residuals are consistent with the normality assumption. In addition, the CUSUM and CUSUM of squares tests provide evidence of structural stability, indicating that the ARDL model is free from detectable structural breaks at conventional significance levels. Collectively, these diagnostic outcomes validate the methodological robustness of the study and strengthen confidence in its conclusions regarding the adverse impact of exchange rate volatility on FDI in Egypt during 2001–2024.

Whereas the empirical results of this study carry important theoretical and practical implications. The confirmation of a negative short-run and long-run relationship between exchange rate volatility and FDI inflows reinforces classical economic theory on the adverse role of uncertainty in investment decisions. From a policy perspective, the results highlight the urgent need for Egyptian authorities to adapt exchange rate stabilization measures as part of broader macroeconomic reforms to enhance the investment climate. Practically, the EGARCH–ARDL framework employed here provides investors and policymakers with a robust tool for modeling volatility clusters and anticipating risk, thereby informing both investment strategies and financial regulation. By demonstrating the critical link between exchange rate stability and foreign investment, this study contributes to guiding policies that can promote sustainable economic growth in Egypt.

Therefore, the study has some limitations. The scope of variables analysis focuses primarily on short- and long-run volatility dynamics, without incorporating other macroeconomic determinants such as trade openness, interest rates, or political risk. The evidence suggests that aggregate FDI inflows respond differently across sectors, supporting the importance of disaggregated analysis in empirical research. Acknowledging these sectoral variations helps explain inconsistencies in prior studies and mitigates the risk of aggregation bias. Although this study addresses key methodological gaps, its scope remains limited to short- and long-run volatility dynamics, leaving scope for future research to incorporate additional macroeconomic determinants.

5. Conclusion

This study provides empirical evidence on the impact of exchange rate volatility on FDI inflows in Egypt over the period 2001–2024, employing the EGARCH–ARDL framework. The findings confirm that VOLREXR exerts a statistically significant negative effect on aggregate FDI inflows in both the short and long run. By addressing the central research question, the analysis demonstrates that heightened currency uncertainty weakens investor confidence. It constrains capital inflows, thereby highlighting the critical importance of exchange rate stability in sustaining investment growth.

The contribution of this research lies in its comprehensive application of the ARDL methodology to Egypt's economy, offering robust empirical evidence that extends theoretical perspectives on the determinants of FDI. By integrating exchange rate volatility with broader macroeconomic factors, the study advances understanding of the complex interplay between uncertainty, growth, and financial development in influencing investment flows. In doing so, it highlights uncertainty as a pivotal determinant of investment behavior and provides novel insights into the dynamic relationship between exchange rate movements and FDI.

Beyond its methodological contribution, the findings carry significant implications for theory, practice, and policy. They reinforce the importance of macroeconomic stability, sustained growth, and financial market development as strategic levers for attracting foreign capital. For policymakers, the results emphasize that mitigating exchange rate volatility, developing capital market expansion, and strategically managing EXDBTG are essential measures for strengthening Egypt's investment climate. Ultimately, these strategies are vital for enhancing long-term economic resilience and advancing sustainable development.

This study supported that macroeconomic determinants play a decisive role in attracting FDI inflows into Egypt. Market size, proxied by GDP, emerges as a significant driver of investment in both the short and long run, supporting the market-seeking FDI hypothesis and aligning with global and Egyptian empirical evidence. Inflation, measured by the CPI, also demonstrates a positive and statistically significant relationship with FDI, a finding that challenges conventional expectations but reflects context-specific dynamics in Egypt. In addition, MCAP shows a strong positive association with FDI, supporting the importance of deep and liquid capital markets in attracting international investment. Conversely, EXDBTG burdens initially deter FDI in the short run but contribute positively in the long run when strategically managed, highlighting the dual role of debt as both a source of risk and a mechanism for financing development projects. Collectively, these findings reinforce the robustness of the ARDL and EGARCH framework and provide valuable insights for policymakers seeking to enhance Egypt's resilience to volatility and attract sustainable foreign investment.

While the study provides valuable empirical evidence about the impact of exchange rate volatility on FDI in Egypt, 2001-2024, it also acknowledges limitations related to variable scope and aggregation. Future research should explore sector-specific investment responses, incorporate additional macroeconomic determinants, and extend comparative analysis to other emerging economies. Such work would deepen understanding of how structural factors interact with volatility and inform more comprehensive policy strategies. Ultimately, this study affirms that exchange rate stability, market growth, and sound financial management are central to promoting investor confidence and supporting sustainable economic development in Egypt. By clarifying the link between volatility, macroeconomic fundamentals, and foreign investment, it provides both theoretical reinforcement and practical guidance for shaping policies that advance long-term growth.

Overall, the study demonstrates that VOLREXR is a critical determinant of FDI in Egypt. Stabilizing exchange rates, ensuring macroeconomic stability, and strengthening financial markets are essential policy measures to attract and sustain foreign investment.

6. Recommendations

Based on the study findings and within the scope of its objectives, several recommendations can be proposed to improve the FDI climate in Egypt. Several policy implications are as follows:

6.1 Strengthening Exchange Rate Stability

Action: The CBE should intervene as necessary to smooth out excessive fluctuations or provide investment guarantees to strategic sectors during periods of exchange rate volatility, to stabilize expectations, particularly for risk-averse investors, necessitating reinforced policy credibility, in addition to publishing regular updates on exchange rate policy and macroeconomic outlook to avoid excessive real exchange rate fluctuations that discourage FDI, especially in non-oil sectors.

Policy strategy: policymakers should prioritize exchange rate stabilization and adopt a flexible but well-managed exchange rate regime and intervene in currency markets during speculative shocks, also improve FX reserves to support market confidence, through credible coordinating fiscal, monetary, and macroeconomic frameworks to secure financial, macroeconomic and exchange rate stability, the existence of a consistent and sustainable comprehensive economic policy framework is a fundamental requirement for attracting FDI, reduces uncertainty, enhances investor confidence, and provides a predictable environment in which long-run investment decisions can be made.

6.2 Stimulating Economic Growth through Macroeconomic Policy

Action: Strengthen macroeconomic fundamentals to expand market size and improve investor confidence; on the other hand, FDI flows towards underdeveloped sectors beyond petroleum and attract investment in sectors aligned with national development goals, in addition to helping investors hedge against exchange rate risks.

Strategy: Policymakers should adopt appropriate comprehensive macroeconomic strategies that strengthen the economy and accelerate GDP growth resilience by expanding public investment in infrastructure and productivity-enhancing sectors such as transport and energy, while simultaneously promoting private sector participation through deregulation and fiscal incentives and maintaining macro-fiscal discipline to reduce debt and inflationary pressures; together, these measures will generate stronger growth prospects and market expansion, thereby attracting long-term FDI, particularly in service-oriented and knowledge-based industries, reflecting a shift away from resource-dependent investment sectors.

6.3 Inflation Targeting as a National Priority

Action: Quantify the relationship between inflation volatility, exchange rate movements, and FDI inflows using econometric models to test whether inflation targeting as a central objective of macroeconomic policy reduces uncertainty and improves investor confidence in Egypt's context.

Policy strategy: Policymakers should maintain a clear and transparent inflation target in addition to using interest rate policy to anchor inflation expectations; furthermore, coordinate with fiscal policy to reduce budget deficits, which affect inflation and currency volatility. This requires addressing structural and real distortions that intensify inflationary pressures, and establishing a clear and stable target for a reduced inflation rate will enhance competitiveness and contain expected inflation, thereby reinforcing macroeconomic credibility and supporting sustainable investment flows.

6.4 The Egyptian Capital Market Authority

Action: Strengthen the CMA's regulatory role in stabilizing and developing capital markets and channeling foreign capital into long-term productive investment.

Strategy: Policymakers should adopt a proactive, investor-friendly regulatory framework that enhances transparency, streamlines licensing and listing procedures, and ensures robust investor protection. This includes harmonizing capital market regulations with international best practices to reduce perceived risk. Establishing a dedicated unit to facilitate foreign institutional investor onboarding. Analyzing the behavior of foreign investors to design policies that discourage speculative activity while encouraging the conversion of short-run capital movements into long-term investment commitments, thereby stabilizing the investment environment and enhancing the capacity to attract sustainable FDI.

6.5 Strategic External Debt Management

Action: Adopt a medium-term debt strategy that balances EXDBTG accumulation with repayment capacity, focusing on cost-risk tradeoffs, maturity lengthening, and currency composition, and enhance risk metrics and thresholds, monitor debt service-to-exports, average time to refix rates, and FX-denominated share; set prudential limits and publish annual debt sustainability analyses.

Strategy: policymakers must align debt with FDI and FX earnings and also prioritize borrowing for sector projects that generate foreign exchange (ports, export manufacturing, renewables) and avoid FX-denominated debt for purely domestic-revenue projects. Moreover, liability management operations conduct buybacks, exchanges, and reprofiling to smooth maturities; build precautionary buffers (reserves, contingent credit lines) to manage rollover risk during shocks. This impact strengthens debt sustainability, preserves macro credibility, lowers country risk, and reduces the volatility premium demanded by FDI, especially in non-extractive sectors.

6.6 Establishing a Dedicated Office for Foreign Direct Investment Promotion

Action: Establish a specialized FDI facilitation unit under GAFI and create a new office within GAFI's existing structure dedicated exclusively to foreign investor services, and establish a specialized institution under GAFI responsible for planning, organizing, and directing foreign investments, marketing projects domestically and internationally, and setting clear procedures for maximizing FDI returns.

Strategy: policymakers must streamline the investment process by institutionalizing specialized support, thereby reducing administrative delays and enhancing investor confidence in Egypt's regulatory environment. ensure consistent promotion messages and targeted investor engagement abroad, leveraging diplomatic channels to attract strategic FDI inflows and strengthen Egypt's global investment profile. reduce bureaucratic complexity and transaction costs by offering centralized regulatory guidance, thereby creating a more transparent and efficient environment for foreign investors.

6.7 Organizing an International Conference Promoting Investment in Egypt

International Investment Conference and Promotion Strategy:

Action: Organize a high-profile international conference to showcase Egypt's economic transformations and launch a comprehensive investment promotion strategy.

Strategy: Strengthen Egypt's global visibility, institutionalize investment promotion capacity, and ensure that FDI inflows are systematically aligned with national priorities, thereby enhancing investor confidence and competitiveness in the global investment landscape.

Author Contributions

Conceptualization, E.S.K.A. and A.S.E.S.; methodology, E.S.K.A.; software, A.S.E.S.; validation, E.S.K.A., A.S.E.S., and M.E.; formal analysis, E.S.K.A.; investigation, A.S.E.S.; resources, M.E.; data curation, A.S.E.S.; writing—original draft preparation, E.S.K.A.; writing—review and editing, A.S.E.S. and M.E.; visualization, A.S.E.S.; supervision, E.S.K.A.; project administration, E.S.K.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon reasonable request. All macroeconomic and financial variables employed in the analysis were obtained from publicly accessible databases, including the World Bank statistics and the CBE reports. No proprietary or restricted datasets

were used.

Acknowledgements

The authors wish to acknowledge the valuable contributions of the Department of Finance, College of Management and Technology, Arab Academy for Science, Technology and Maritime Transportation, Alexandria, Egypt. In addition, we extend our appreciation to those who provided constructive feedback on earlier drafts of the manuscript, which helped improve its clarity, analytical depth, and methodological strength.

Conflicts of Interest

The authors declare no conflicts of interest.

References

- Abdalla, S. Z. S. (2012). Modelling exchange rate volatility using GARCH models: Empirical evidence from Arab countries. *Int. J. Econ. Finance*, 4(3), 216–229. <https://doi.org/10.5539/ijef.v4n3p216>.
- Abdallah, R. T. (2023). The impact of the exchange rate on foreign direct investment in Egypt. *Sci. J. Bus. Environ. Stud.*, 14(3), 3707–3737. <https://doi.org/10.21608/jces.2023.323250>.
- Abdel Aziz, A. & Al-Ajrawi, A. (2023). The impact of the exchange rate on foreign direct investment in the Iraqi agricultural sector for the period 1990–2021. *Tikrit J. Agric. Sci.*, 23(1), 189–201. <https://doi.org/10.25130/tjas.23.1.24>.
- Abdelgany, M. (2020). The impact of FDI on exports in Egypt. *J. Econ. Polit.*, 5(6), 1–37. <https://doi.org/10.21608/JOCU.2020.46058.1066>.
- Adokwe, E. I., Agu, A. O., & Maduka, A. C. (2019). Exchange rate volatility and foreign direct investment: The Nigerian experience. *J. Bus. Econ. Policy*, 6(4), 78–87. <https://doi.org/10.30845/jbep.v6n4p10>.
- Akinlo, A. E. & Onatunji, O. G. (2020). Exchange rate volatility and domestic investment: Evidence from twelve ECOWAS countries. *Afr. J. Econ. Rev.*, 8(2), 176–189.
- Akinlo, A. E. & Onatunji, O. G. (2021). Exchange rate volatility and foreign direct investment in selected West African countries. *Int. J. Bus. Finance Res.*, 15(1), 77–88.
- Al-Ghiash, M. M. I. (2022). Optimal exchange rate limits that attract foreign direct investment to Egypt. *J. Polit. Econ.*, 15(14). <https://doi.org/10.21608/jocu.2022.128278.1167>.
- Alnaa, S. E. & Ahiakpor, F. (2020). Exchange rate volatility and foreign direct investment. *Res. Appl. Econ.*, 12(3). <https://doi.org/10.5296/rae.v12i3.17737>.
- Al-Sayed, Y. I. K. (2020). A standard model for the impact of the exchange rate on foreign direct investment in Egypt for the period 1977–2017. *J. Strateg. Stud. Disasters Opportunity Manag.*, 2(4), 129–158. <https://search.mandumah.com/Record/1066657>
- Ambala, R. N. & Amewu, G. (2022). Linear and non-linear ARDL estimation of financial innovation and economic growth in Ghana. *J. Bus. Socio-Econ. Dev.*, 3(1), 36–49. <https://doi.org/10.1108/JBSED-09-2021-0128>.
- Arvin, M. B., Pradhan, R. P., & Nair, M. S. (2021). Are there links between institutional quality, government expenditure, tax revenue, and economic growth? Evidence from low-income and lower middle-income countries. *Econ. Anal. Policy*, 70, 468–489. <https://doi.org/10.1016/j.eap.2021.03.011>.
- Asamoah, M. E., Alagidede, I. P., & Adu, F. (2022). Exchange rate uncertainty and foreign direct investment in Africa: Does financial development matter? *Rev. Dev. Econ.*, 26(2), 878–898. <https://doi.org/10.1111/rode.12858>.
- Asmae, A. & Ahmed, B. (2019). Impact of the exchange rate and price volatility on FDI inflows: Case of Morocco and Turkey. *Appl. Econ. Finance*, 6(3), 87–104. <https://doi.org/10.11114/aef.v6i3.4218>.
- Balaban, S., Živkov, D., & Milenković, I. (2019). Impact of an unexplained component of real exchange rate volatility on FDI: Evidence from transition countries. *Econ. Syst.*, 43(3–4), 100719. <https://doi.org/10.1016/j.ecosys.2019.100719>.
- Ben-Obi, O. A., Olaniyi, O., Okoroafor, D. O. K., & Obi, B. (2025). Asymmetric effect of exchange rate volatility on foreign direct investment, inflation, and balance of trade in Nigeria. *Am. J. Ind. Bus. Manag.*, 15, 681–698. <https://doi.org/10.4236/ajibm.2025.155033>.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *J. Econom.*, 31(3), 307–327. [https://doi.org/10.1016/0304-4076\(86\)90063-1](https://doi.org/10.1016/0304-4076(86)90063-1).
- Breusch, T. S. (1978). Testing for autocorrelation in dynamic linear models. *Aust. Econ. Pap.*, 17(31), 334. <https://doi.org/10.1111/j.1467-8454.1978.tb00635.x>.
- Ejaz, M. & Azam, A. (2024). Impact of capital inflows on exchange rate in developing economies: A panel data analysis. *Pak. J. Soc. Sci.*, 44(1), 41–52. <https://doi.org/10.5281/zenodo.10512954>.

- El Mahdy, A., Toraeh, N., & Mohamed, R. O. (2022). External debt and economic growth in Egypt. *Sci. J. Res. Commer. Stud.*, 36(4), 153–177. https://sjrbs.journals.ekb.eg/article_274125_a30ea2c0f5ad78d7c5d7aa3090ee4aeb.pdf
- El Rhadbane, A. B. & El Moudden, A. (2022). COVID-19 pandemic, real effective exchange rate and foreign direct investment inflows: Evidence from Morocco, Turkey and Egypt. *Int. J. Appl. Econ. Finance Account.*, 12(2), 38–51. <https://doi.org/10.33094/ijaefa.v12i2.543>.
- Elagouza, E. (2023). The distributional effects of liberalizing the exchange rate in the Egyptian economy during 1990–2018. *Arab J. Adm.*, 43(4), 245–278. <https://doi.org/10.21608/AJA.2021.65107.1054>.
- Elbadry, E. & Mandour, A. (2020). The effect of exchange rate on FDI flows using the distributed time gap autoregression model (ARDL) in the Egyptian economy during the period 1990–2018. *J. Bus. Res.*, 42(2), 120–152. <https://search.mandumah.com/Record/1082364>
- Elmoghbar, M. S. (2025). The effect of exchange rate movements on net inflows of foreign direct investment during the period 1974–2023: An empirical study on Egypt. *J. Alexand. Univ. Adm. Sci.*, 62(2), 97–141. <https://doi.org/10.21608/acj.2025.416425>.
- El-Sayed, Y. I. K. (2020). The effect of exchange rate on foreign direct investment in Egypt: An econometric model. *Ain Shams Scholar*, 2(5), 105–128. <https://research.asu.edu.eg/handle/123456789/177504>
- Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007. <https://doi.org/10.2307/1912773>.
- Ezekiel, O. A. & Temidayo, O. A. (2023). Does financial deepening matter in the nexus between exchange rate volatility and foreign investment? Insight from Nigeria. *Afr. J. Econ. Rev.*, 11(1), 33–49.
- Fadl, N. & Ghoneim, H. (2020). Impact of foreign exchange rate on foreign direct investment: Egypt case. *Rev. Lareiid*, 2020, 138–159.
- Faizi, M. N., Khan, M. I., Mohammad, A., & Aghaz, Q. S. (2024). An empirical analysis of exchange rate volatility and foreign investment inflows from the post-reform period to 2020 in India. *Eur. Econ. Lett.*, 14(1), 127–133. <https://doi.org/10.52783/eel.v14i1.1006>.
- Fatima, E., Asif, M., Sharma, R. B., & Chaudhary, A. (2023). Exchange rate volatility and its impact on FDI inflows in India using Maki cointegration approach. In *Digitalisation: Opportunities and Challenges for Business* (pp. 392–406). Springer. https://doi.org/10.1007/978-3-031-26953-0_37.
- Frenzel Baudisch, C. (2018). *Sectoral FDI and the real exchange rate: The role of financial development* (No. 28-2018). <https://hdl.handle.net/10419/200684>
- Godfrey, L. G. (1978). Testing against general autoregressive and moving average error models when the regressors include lagged dependent variables. *Econometrica*, 46(6), 1293–1301. <https://doi.org/10.2307/1913829>.
- Hamida, A. (2024). The impact of exchange rate volatility on foreign direct investment: ARDL testing bound. *Int. J. Islam. Bus. Manag.*, 8(1), 12–17. <https://doi.org/10.46281/ijibm.v8i1.2222>.
- He, Y. (2018). A study on the impact of real exchange rate volatility of RMB on China's foreign direct investment to Japan. *East Asian J. Bus. Econ.*, 6(3), 24–36. <https://doi.org/10.20498/eajbe.2018.6.3.24>.
- Hend, M. A. E., Ahmed, A. Z. & Hosni, H. M. (2021). Effects of currency devaluation on foreign investments in Egypt. *Acad. J. Res. Sci. Publ.*, 29, 383–406. <https://www.ajrsp.com/vol/issue29/%D8%A2%D8%AB%D8%A7%D8%B1%20%D8%AA%D8%AE%D9%81%D9%8A%D8%B6%20%D9%82%D9%8A%D9%85%D8%A9%20%D8%A7%D9%84%D8%B9%D9%85%D9%84%D8%A9%20%D8%B9%D9%84%D9%89%20%D8%A7%D9%84%D8%A7%D8%B3%D8%AA%D8%AB%D9%85%D8%A7%D8%B1%D8%A7%D8%AA%20%D8%A7%D9%84%D8%A3%D8%AC%D9%86%D8%A8%D9%8A%D8%A9%20%D9%81%D9%8A%20%D9%85%D8%B5%D8%B1.pdf>
- Hniya, S., Boubker, A., Mrad, F., & Nafti, S. (2021). The impact of real exchange rate volatility on foreign direct investment inflows in Tunisia. *Int. J. Econ. Financ. Issues*, 11(5), 52–67. <https://doi.org/10.32479/ijefi.11882>.
- Huong, T. T. X., Nguyen, M.-L. T., & Lien, N. T. K. (2020). An empirical study of the real effective exchange rate and foreign direct investment in Vietnam. *Invest. Manag. Financ. Innov.*, 17(4), 1–13. [https://doi.org/10.21511/imfi.17\(4\).2020.01](https://doi.org/10.21511/imfi.17(4).2020.01).
- Imamboccus, R., Seetanah, B., Nunkoo, R., & Jaffur, Z. K. (2024). The impact of exchange rate and exchange rate volatility on tourism demand using disaggregated data. *Int. J. Tour. Res.*, 26(2), e2640. <https://doi.org/10.1002/jtr.2640>.
- Jamal, A. & Bhat, G. M. (2022). Disentangling the nexus between exchange rate volatility, exports, and FDI: Empirical evidence from the Indian economy. *Glob. J. Emerg. Mark. Econ.*, 15(3), 449–472. <https://doi.org/10.1177/09749101221108788>.
- Jannat, Z. (2020). The impact of exchange rate volatility on foreign direct investment inflow: Evidence from South Asia. *Int. J. Finance Insur. Risk Manag.*, 10(3), 101–116.
- Jarque, C. M. & Bera, A. K. (1980). Efficient tests for normality, homoscedasticity, and serial independence of regression residuals. *Econ. Lett.*, 6(3), 255–259. [https://doi.org/10.1016/0165-1765\(80\)90024-5](https://doi.org/10.1016/0165-1765(80)90024-5).

- Joel, O. & Jeffrey, O. E. (2023). Exchange rate volatility and foreign investment growth in Nigeria. *Afr. Dev. Finance J.*, 6(1), 105–123.
- Jokosenumi, S. O. & Adesete, A. A. (2018). Modelling the effect of stock market volatility and exchange rate volatility on foreign direct investment in Nigeria: A new framework approach. *Asian Econ. Financ. Rev.*, 8(12), 1482–1505.
- Kalu, G. C. (2024). Impact of exchange rate volatility on foreign direct investment in Nigeria (1981–2018). *Br. Int. J. Appl. Econ. Finance Account.*, 8(2). <https://aspjournals.net/Journals/index.php/bijaefa/article/view/580/651>
- Kaya, A. I. & Erden, L. (2022). Capital-flow volatility in emerging markets: A panel GARCH approach. *Int. Finance*, 26(2), 172–188. <https://doi.org/10.1111/infi.12427>.
- Kiliçarslan, Z. (2018). The relationship between exchange rate volatility and foreign direct investment in Turkey: Toda and Yamamoto causality analysis. *Int. J. Econ. Financ. Issues*, 8(4), 61–67.
- Lajevardi, H. & Chowdhury, M. (2024). How does the exchange rate and its volatility influence FDI to Canada? A disaggregated analysis. *J. Risk Financ. Manag.*, 17(2), 88. <https://doi.org/10.3390/jrfm17020088>.
- Latief, R. & Lefen, L. (2018). The effect of exchange rate volatility on international trade and foreign direct investment (FDI) in developing countries along “One Belt and One Road.” *Int. J. Financ. Stud.*, 6(4), 86. <https://doi.org/10.3390/ijfs6040086>.
- Lin, C.-C. & Chen, K.-M. (2022). Market competition, exchange rate uncertainty, and foreign direct investment. *Rev. Dev. Econ.*, 26(1), 405–422. <https://doi.org/10.1111/rode.12813>.
- Mackton, W. S., Oondo, A., & Nyongesa, D. (2018). Real effective exchange rate volatility and its impact on foreign direct investment in Kenya. *Asian J. Econ. Bus. Account.*, 6(4), 1–20. <https://doi.org/10.9734/AJEBA/2018/38008>.
- Marwa, S. A., Hatem, A. E., & Mervat, M. A. (2024). The relationship between the exchange rate and the foreign direct investment measuring model in Egypt (1990–2022). *Sci. J. Bus.*, 52(1), 239–284. <https://doi.org/10.21608/sjsc.2023.234017.1345>.
- Moraghen, W., Seetanah, B., & Sookia, N. (2021). Impact of exchange rate and exchange rate volatility on foreign direct investment inflow for Mauritius: A dynamic time series approach. *Afr. Dev. Rev.*, 33(4), 581–591. <https://doi.org/10.1111/1467-8268.12596>.
- Muhammad, M. A. & Abdullahi, K. (2020). Impact of external debt servicing on economic growth in Nigeria: An ARDL approach. *Int. J. Bus. Technopreneurship*, 10(2), 257–268.
- Musyoka, N. (2018). *The effect of real interest rate, inflation, exchange rate and competitiveness on foreign direct investment in Kenya* [Mastersthesis, Kenyatta University, Kenya]. <https://ir-library.ku.ac.ke/server/api/core/bitstreams/7f50cf17-b4b9-422f-b352-bc39b1846c6a/content>
- Nadine, A.-E. A., Ashraf, S., & Nagia, R. (2021). The impact of relative exchange rate volatility and other multidimensional determinants on FDI in Egypt. *Am. J. Ind. Bus. Manag.*, 11, 1163–1197. <https://doi.org/10.4236/ajibm.2021.1112071>.
- Nduka, A. J., Nwankwor, V. A., & Egungwu, I. (2024). The effect of foreign exchange fluctuations on foreign direct investment inflows in Nigeria. *J. Emerg. Trends Manag. Sci. Entrep.*, 6(1), 1–25. <https://www.jetmase.com/index.php/jetmase/article/view/80>
- Nelson, D. B. (1991). Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2), 347–370. <https://doi.org/10.2307/2938260>.
- Nguyen, T. T., Nasir, M. A., & Vo, X. V. (2022). Exchange rate dynamics of emerging and developing economies: Not all capital flows are alike. *Int. J. Finance Econ.*, 29(1), 1115–1124. <https://doi.org/10.1002/ijfe.2724>.
- Okonkwo, J. J., Osakwe, C. I., & Nwadike, E. C. (2021). Exchange rate and foreign direct investment in Nigeria 1981–2018. *Int. J. Acad. Res. Bus. Soc. Sci.*, 11(1), 213–232. <https://doi.org/10.6007/IJARBS/v11-i1/8466>.
- Oladeji, B. & Musa, M. (2022). Effect of exchange rate volatility on foreign direct investment in Nigeria [Preprint]. *ScienceOpen*. <https://doi.org/10.14293/S2199-1006.1.SOR-PPBL3HY.v1>.
- Ozigbo, S. A. & Anuya, M. O. (2023). Exchange rate dynamics and foreign direct investment in Nigeria. *J. Xidian Univ.*, 17(7), 1047–1061. <https://doi.org/10.37896/jxu17.7/086>.
- Qamruzzaman, M., Mehta, A. M., Khalid, R., Serfraz, A., & Saleem, H. (2021). Symmetric and asymmetric effects of financial innovation and FDI on exchange rate volatility: Evidence from South Asian countries. *J. Asian Finance Econ. Bus.*, 8(1), 23–36. <https://doi.org/10.13106/jafeb.2021.vol8.no1.023>.
- Rahman, M. M. & Kashem, M. A. (2017). Exploring the linkage between financial development and economic growth: The ARDL bounds testing approach [Preprint]. *Research Square*, 1–21.
- Rasheed, R. A., Kamsin, A., & Abdullah, N. A. (2020). Challenges in the online component of blended learning: A systematic review. *Comput. Educ.*, 144, 103701. <https://doi.org/10.1016/j.compedu.2019.103701>.
- Salmerón-Gómez, R., García-García, C. B. & García-Pérez, J. (2025). A redefined variance inflation factor: Overcoming the limitations of the variance inflation factor. *Comput. Econ.*, 65, 337–363. <https://doi.org/10.1007/s10614-024-10575-8>.
- Shabbir, A., Abbas, S., Javed, I., & Nisar, N. (2025). The role of tax policies, inflation, trade openness, and

- exchange rate volatility on the foreign investment decisions in Pakistan. *Indus J. Soc. Sci.*, 3(2).
- Shafique, A., Hassan, M. U., Shahzad, A., Ali, Q. M., & Saqlain, M. (2022). Exchange rate volatility and its relationship with macroeconomic variables in Pakistan. *Bull. Bus. Econ.*, 11(1), 121–131. <https://doi.org/10.5281/zenodo.6412579>.
- Sultana, M., Rahman, M. H., & Zimon, G. (2024). The impact of external debt and exchange rate on foreign direct investment in emerging investment markets: New evidence using a PMG-ARDL panel data analysis. *Cogent Econ. Finance*, 12(1), 2362780. <https://doi.org/10.1080/23322039.2024.2362780>.
- Thujiyanthan, P. (2021). The impact of exchange rate and exchange rate volatility on foreign direct investment: An econometric investigation in Sri Lanka. *Asian J. Manag. Sci.*, 10(2), 41–51. <https://doi.org/10.51983/ajms-2021.10.2.2928>.
- Udoinyang, N. & Udoinyang, N. (2024). The influence of exchange rate fluctuation on foreign direct investment in Nigeria (1986–2022). *J. Econ. Innov. Manag. Entrep.*, 2(1), 16–33. <https://doi.org/10.59652/jeime.v2i1.139>.
- Zaharum, Z., Md Isa, M. A., Latif, R. A., & Md Isa, M. F. (2024). Determinants of foreign direct investment in Malaysia. *Insight J.*, 11(1), 86–96. <https://ir.uitm.edu.my/id/eprint/95213>
- Zongo, E., Dama, E., Yenyetou, D., Muhigwa, M., Nikiema, A., Dahourou, G. A., & Ouedraogo, A.-S. (2024). On-site evaluation as external quality assessment of microbiology laboratories involved in sentinel laboratory-based antimicrobial resistance surveillance network in Burkina Faso. *Antimicrob. Resist. Infect. Control*, 13(3). <https://doi.org/10.1186/s13756-023-01362-5>.